Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c,\ b = d$$

Powers

$$i^4 = 1 \qquad i^{4q+r} = i^r$$

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Worked Solutions **Complex Numbers**

Q001 Written

State the real and imaginary parts of $3 - 2i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(3,\ -2)$.

Final answer

$(3,\ -2)$

Q002 Written

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(4/3, -\sqrt{5}/3)$.

Final answer

$(4/3, -\sqrt{5}/3)$

Worked Solutions **Complex Numbers**

Q003 Written

State the real and imaginary parts of -8.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-8,\ 0)$.

Final answer

$(-8,\ 0)$

Q004 Written

State the real and imaginary parts of $2i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ 2)$.

Final answer

$(0,\ 2)$

Worked Solutions **Complex Numbers**

Q005 Written

Find the real values of x and y satisfying $(2+i)x+(1+i)y=2+3i, x, y real$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=-1, y=4$.

Final answer

$x=-1, y=4$

Q006 Written

State the real and imaginary parts of $-3+5i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-3,\ 5)$.

Final answer

$(-3,\ 5)$

Worked Solutions **Complex Numbers**

Q007 Written

State the real and imaginary parts of $(-1-\sqrt{3}i)/2$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-1/2, -\sqrt{3}/2)$.

Final answer

$(-1/2, -\sqrt{3}/2)$

Q008 Written

State the real and imaginary parts of 1.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1,\ 0)$.

Final answer

$(1,\ 0)$

Worked Solutions **Complex Numbers**

Q009 Written

State the real and imaginary parts of $3i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ 3)$.

Final answer

$(0,\ 3)$

Q010 Written

Find the real values of x and y satisfying $(x - 6) + (x + 3y)i = 0, x, yreal$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 6, y = -2$.

Final answer

$x = 6, y = -2$

Worked Solutions **Complex Numbers**

Q011 Written

Find the real values of x and y satisfying $(3+2i)x+(4-i)y=6-7i, x, yreal$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=-2, y=3$.

Final answer

$x=-2, y=3$

Q012 Written

State the real and imaginary parts of $2+5i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2,\ 5)$.

Final answer

$(2,\ 5)$

Worked Solutions **Complex Numbers**

Q013 **Written**

State the real and imaginary parts of $-25 + 13i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-25,\ 13)$.

Final answer

$(-25,\ 13)$

Q014 **Written**

State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2/3, -\sqrt{5}/3)$.

Final answer

$(2/3, -\sqrt{5}/3)$

Worked Solutions **Complex Numbers**

Q015 **Written**

State the real and imaginary parts of $(7-\sqrt{2}i)/2$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(7/2, -\sqrt{2}/2)$.

Final answer

$(7/2, -\sqrt{2}/2)$

Q016 **Written**

State the real and imaginary parts of -4.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-4,\ 0)$.

Final answer

$(-4,\ 0)$

Worked Solutions **Complex Numbers**

Q017 Written

State the real and imaginary parts of $1/4$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1/4,\ 0)$.

Final answer

$(1/4,\ 0)$

Q018 Written

State the real and imaginary parts of $\sqrt{3}i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, \sqrt{3})$.

Final answer

$(0, \sqrt{3})$

Worked Solutions **Complex Numbers**

Q019 Written

State the real and imaginary parts of $-i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ -1)$.

Final answer

$(0,\ -1)$

Q020 Written

Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = 2$.

Final answer

$x = 3, y = 2$

Worked Solutions **Complex Numbers**

Q021 **Written**

Find the real values of x and y satisfying $(x+1)-5i=4+(2y+1)i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=3, y=-3$.

Final answer

$x=3, y=-3$

Q022 **Written**

Find the real values of x and y satisfying $(x-2)+(x+y)i=0$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=2, y=-2$.

Final answer

$x=2, y=-2$

Worked Solutions **Complex Numbers**

Q023 Written

Find the real values of x and y satisfying $(3 + 2i)x + (1 - i)y = 7 + 3i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = 1$.

Final answer

$x = 2, y = 1$

Q024 Written

Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 1/2, y = 3/2$.

Final answer

$x = 1/2, y = 3/2$

Worked Solutions **Complex Numbers**

Q025 **Written**

Solve the complex-number equation and find the exact real values of x and y: $(1 + 2i)(x + i) = x(y + i)$.

Solution

1. Expand: $((1 + 2i)(x + i) = x - 2 + (1 + 2x)i)$.
2. Equate coefficients with (xy+xi): $(x - 2 = xy)$ and $(1 + 2x = x)$.
3. Solving gives $x = -1, y = 3$.

Final answer

$x = -1, y = 3$

Q026 **Written**

Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 7i$.

Final answer

$1 - 7i$

Worked Solutions **Complex Numbers**

Q027 **Written**

Calculate $(i-2)^2$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $3-4i$.

Final answer

$3-4i$

Q028 **Written**

Calculate $i^6+i^{27}+i^{17}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -1.

Final answer

-1

Worked Solutions **Complex Numbers**

Q029 Written

Calculate $(4+6i)+(-3+2i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1+8i$.

Final answer

$1+8i$

Q030 Written

Calculate $(2+3i)(1-2i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $8-i$.

Final answer

$8-i$

Worked Solutions **Complex Numbers**

Q031 Written

Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 0.

Final answer

0

Q032 Written

Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $4 - i$.

Final answer

$4 - i$

Worked Solutions **Complex Numbers**

Q033 Written

Calculate $(-4+3i)+(5-6i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1-3i$.

Final answer

$1-3i$

Q034 Written

Calculate $(3-3i)-(5-3i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -2.

Final answer

-2

Worked Solutions **Complex Numbers**

Q035 **Written**

Calculate $(\sqrt{2}+3i)(\sqrt{2}+i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-1+4\sqrt{2}i$.

Final answer

$-1+4\sqrt{2}i$

Q036 **Written**

Calculate $(2+3i)+(5+8i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $7+11i$.

Final answer

$7+11i$

Worked Solutions **Complex Numbers**

Q037 Written

Calculate $(2/5 - 1/2i) - (1/3 + 3/7i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1/15 - 13/14i$.

Final answer

$1/15 - 13/14i$

Q038 Written

Calculate $(2i - 1)^2$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-3 - 4i$.

Final answer

$-3 - 4i$

Worked Solutions **Complex Numbers**

Q039 Written

Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -3.

Final answer

-3

Q040 Written

Calculate $(2 + i)^3 + (2 - i)^3$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 4.

Final answer

4

Worked Solutions **Complex Numbers**

Q041 Written

Calculate $i^5 + i^3 + i$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: i.

Final answer

i

Q042 Written

Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -1.

Final answer

-1

Worked Solutions **Complex Numbers**

Q043 **Written**

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-i$.

Final answer

$-i$

Q044 **Written**

Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2 i^{40}$.

Solution

1. Use ($i^{40} = 1$) and (($x - i)(x + i) = x^2 + 1$).
2. The equation becomes ($7x(x^2 + 1) = 14x^2$), so ($7x(x - 1)^2 = 0$).
3. Therefore $x = 0$ or $x = 1$.

Final answer

$x = 0$ or $x = 1$

Worked Solutions **Complex Numbers**

Q045 **Written**

State the complex conjugate of $2 + 3i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $2 - 3i$.

Final answer

$2 - 3i$

Q046 **Written**

State the complex conjugate of $7 - i\sqrt{11}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $7 + i\sqrt{11}$.

Final answer

$7 + i\sqrt{11}$

Worked Solutions **Complex Numbers**

Q047 **Written**

State the complex conjugate of 6.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 6.

Final answer

6

Q048 **Written**

State the complex conjugate of $-5i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $5i$.

Final answer

$5i$

Worked Solutions **Complex Numbers**

Q049 Written

Calculate $(2+i)/(2-i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5+4/5i$.

Final answer

$3/5+4/5i$

Q050 Written

Calculate $(3-i)/(2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1/2-3/2i$.

Final answer

$-1/2-3/2i$

Worked Solutions **Complex Numbers**

Q051 **Written**

State the complex conjugate of $1 - 4i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $1 + 4i$.

Final answer

$1 + 4i$

Q052 **Written**

State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{2} - i\sqrt{10}$.

Final answer

$\sqrt{2} - i\sqrt{10}$

Worked Solutions **Complex Numbers**

Q053 **Written**

State the complex conjugate of 2.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 2.

Final answer

2

Q054 **Written**

State the complex conjugate of $-8i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8i$.

Final answer

$8i$

Worked Solutions **Complex Numbers**

Q055 **Written**

Calculate $(3+i)/(1+2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1-i$.

Final answer

$1-i$

Q056 **Written**

Calculate $(1-i)/i$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1-i$.

Final answer

$-1-i$

Worked Solutions **Complex Numbers**

Q057 **Written**

State the complex conjugate of $3 + 4i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $3 - 4i$.

Final answer

$3 - 4i$

Q058 **Written**

State the complex conjugate of $9 + 8i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $9 - 8i$.

Final answer

$9 - 8i$

Worked Solutions **Complex Numbers**

Q059 **Written**

State the complex conjugate of $8 + i\sqrt{3}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8 - i\sqrt{3}$.

Final answer

$8 - i\sqrt{3}$

Q060 **Written**

State the complex conjugate of 5.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 5.

Final answer

5

Worked Solutions **Complex Numbers**

Q061 Written

State the complex conjugate of -15.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is -15.

Final answer

-15

Q062 Written

State the complex conjugate of $\sqrt{7}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{7}$.

Final answer

$\sqrt{7}$

Worked Solutions **Complex Numbers**

Q063 **Written**

State the complex conjugate of i.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i$.

Final answer

$-i$

Q064 **Written**

State the complex conjugate of $-50i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $50i$.

Final answer

$50i$

Worked Solutions **Complex Numbers**

Q065 Written

State the complex conjugate of $i\sqrt{6}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i\sqrt{6}$.

Final answer

$-i\sqrt{6}$

Q066 Written

Calculate $(1+i)/(1-i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is i.

Final answer

i

Worked Solutions **Complex Numbers**

Q067 **Written**

Calculate $(3+i)/(2+i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $7/5 - 1/5i$.

Final answer

$7/5 - 1/5i$

Q068 **Written**

Calculate $(\sqrt{5}+i)/(\sqrt{5}-i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2/3 + \sqrt{5}/3i$.

Final answer

$2/3 + \sqrt{5}/3i$

Worked Solutions **Complex Numbers**

Q069 **Written**

Calculate $5i/(1+2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2+i$.

Final answer

$2+i$

Q070 **Written**

Calculate $5i/(4+3i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5+4/5i$.

Final answer

$3/5+4/5i$

Worked Solutions **Complex Numbers**

Q071 **Written**

Calculate $(1-3i)/(3-2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $9/13-7/13i$.

Final answer

$9/13-7/13i$

Q072 **Written**

Calculate $1/i$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-i$.

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q073 **Written**

Calculate $(2 + i)/(3i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1/3 - 2/3i$.

Final answer

$1/3 - 2/3i$

Q074 **Written**

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)}\right)^{2027}$.

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The base simplifies exactly to (displaystyle
rac-2-144i85).
3. Hence the exact result is $((-2 - 144i)/85)^{2027}$; no expansion of the 2027th power is required.

Final answer

$((-2 - 144i)/85)^{2027}$

Worked Solutions **Complex Numbers**

Q075 Written

Express the square roots of $-6intermsofi$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm i\sqrt{6}$.

Final answer

$\pm i\sqrt{6}$

Q076 Written

Express $\sqrt{-12}\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-6\sqrt{2}$.

Final answer

$-6\sqrt{2}$

Worked Solutions **Complex Numbers**

Q077 Written

Express $\sqrt{8}/\sqrt{-18}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-(2/3)i$.

Final answer

$-(2/3)i$

Q078 Written

Express $\sqrt{-7}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $i\sqrt{7}$.

Final answer

$i\sqrt{7}$

Worked Solutions **Complex Numbers**

Q079 Written

Express $\sqrt{-27}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i\sqrt{3}$.

Final answer

$3i\sqrt{3}$

Q080 Written

Express $-\sqrt{-12}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i\sqrt{3}$.

Final answer

$-2i\sqrt{3}$

Worked Solutions **Complex Numbers**

Q081 Written

Express the square roots of -12 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 2i\sqrt{3}$.

Final answer

$\pm 2i\sqrt{3}$

Q082 Written

Express $\sqrt{-3}\,\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-3\sqrt{2}$.

Final answer

$-3\sqrt{2}$

Worked Solutions **Complex Numbers**

Q083 **Written**

Express $\sqrt{27}/\sqrt{-3}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-3i$.

Final answer

$-3i$

Q084 **Written**

Express $(3+\sqrt{-2})^2$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $7+6\sqrt{2}i$.

Final answer

$7+6\sqrt{2}i$

Worked Solutions **Complex Numbers**

Q085 Written

Express $\sqrt{-2}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $i\sqrt{2}$.

Final answer

$i\sqrt{2}$

Q086 Written

Express $\sqrt{-9}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i$.

Final answer

$3i$

Worked Solutions **Complex Numbers**

Q087 **Written**

Express $\sqrt{-45}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i\sqrt{5}$.

Final answer

$3i\sqrt{5}$

Q088 **Written**

Express $-\sqrt{-24}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i\sqrt{6}$.

Final answer

$-2i\sqrt{6}$

Worked Solutions **Complex Numbers**

Q089 **Written**

Express $-\sqrt{-36}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-6i$.

Final answer

$-6i$

Q090 **Written**

Express the square roots of -2 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm i\sqrt{2}$.

Final answer

$\pm i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q091 Written

Express the square roots of -18 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 3i\sqrt{2}$.

Final answer

$\pm 3i\sqrt{2}$

Q092 Written

Express the square roots of -16 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 4i$.

Final answer

$\pm 4i$

Worked Solutions **Complex Numbers**

Q093 Written

Express $\sqrt{-2}\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2\sqrt{3}$.

Final answer

$-2\sqrt{3}$

Q094 Written

Express $\sqrt{-28}\sqrt{-35}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-14\sqrt{5}$.

Final answer

$-14\sqrt{5}$

Worked Solutions **Complex Numbers**

Q095 **Written**

Express $\sqrt{8}/\sqrt{-2}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i$.

Final answer

$-2i$

Q096 **Written**

Express $\sqrt{27}/\sqrt{-12}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-(3/2)i$.

Final answer

$-(3/2)i$

Worked Solutions **Complex Numbers**

Q097 **Written**

Express $\sqrt{-32}+\sqrt{-98}+\sqrt{-8}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $13i\sqrt{2}$.

Final answer

$13i\sqrt{2}$

Q098 **Written**

Express $\sqrt{-98}-\sqrt{-72}+\sqrt{-50}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $6i\sqrt{2}$.

Final answer

$6i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q099 **Written**

Express $(\sqrt{3}+\sqrt{-5})^2$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2+2i\sqrt{15}$.

Final answer

$-2+2i\sqrt{15}$

Q100 **Written**

Express $(2+\sqrt{-3})(2-\sqrt{-3})$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore 7.

Final answer

7

Worked Solutions **Complex Numbers**

Q101 Written

Solve over the complex numbers: $(2x - 8)^2 = -10$.

Solution

1. Take square roots: $(2x - 8 = pm$ isqrt10).
2. Divide by 2 to obtain $x = 4 \pm (\sqrt{10}/2)i$.

Final answer

$x = 4 \pm (\sqrt{10}/2)i$

Q102 MCQ

For $2 + 3i$**, select the ordered pair (real part, imaginary part):**

Solution

1. Simplify $2 + 3i$ using the relevant family rule.
2. The correct value is $(2,\ 3)$.

Correct option: C

Final answer

$(2,\ 3)$

Worked Solutions **Complex Numbers**

Q103 MCQ

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-4 + 5i$ using the relevant family rule.
2. The correct value is $(-4,\ 5)$.

Correct option: D

Final answer

$(-4,\ 5)$

Q104 MCQ

For $7 - 2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $7 - 2i$ using the relevant family rule.
2. The correct value is $(7,\ -2)$.

Correct option: D

Final answer

$(7,\ -2)$

Worked Solutions **Complex Numbers**

Q105 MCQ

For $(1-\sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $(1-\sqrt{2}i)/3$ using the relevant family rule.
2. The correct value is $\left(\frac{1}{3},\ -\frac{\sqrt{2}}{3}\right)$.

Correct option: B

Final answer

$\left(\frac{1}{3},\ -\frac{\sqrt{2}}{3}\right)$

Q106 MCQ

For -6, select the ordered pair (real part, imaginary part):

Solution

1. Simplify -6 using the relevant family rule.
2. The correct value is $(-6,\ 0)$.

Correct option: D

Final answer

$(-6,\ 0)$

Worked Solutions **Complex Numbers**

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4i$ using the relevant family rule.
2. The correct value is $(0,\ 4)$.

Correct option: A

Final answer

$(0,\ 4)$

Q108 MCQ

For $-i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-i$ using the relevant family rule.
2. The correct value is $(0,\ -1)$.

Correct option: B

Final answer

$(0,\ -1)$

Worked Solutions **Complex Numbers**

Q109 MCQ

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{5}i$ using the relevant family rule.
2. The correct value is $\left(0,\ \sqrt{5}\right)$.

Correct option: C

Final answer

$\left(0,\ \sqrt{5}\right)$

Q110 MCQ

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $3/2 + \sqrt{3}i$ using the relevant family rule.
2. The correct value is $\left(\frac{3}{2},\ \sqrt{3}\right)$.

Correct option: B

Final answer

$\left(\frac{3}{2},\ \sqrt{3}\right)$

Worked Solutions **Complex Numbers**

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-1/4 - 2i$ using the relevant family rule.
2. The correct value is $\left(-\frac{1}{4},\, -2\right)$.

Correct option: D

Final answer

$\left(-\frac{1}{4},\, -2\right)$

Q112 MCQ

For $5/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $5/3$ using the relevant family rule.
2. The correct value is $\left(\frac{5}{3},\, 0\right)$.

Correct option: D

Final answer

$\left(\frac{5}{3},\, 0\right)$

Worked Solutions **Complex Numbers**

Q113 *MCQ*

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{7}i$ using the relevant family rule.
2. The correct value is $\left(0,\ -\sqrt{7}\right)$.

Correct option: C

Final answer

$\left(0,\ -\sqrt{7}\right)$

Q114 *MCQ*

For $-9+\sqrt{2}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-9+\sqrt{2}i$ using the relevant family rule.
2. The correct value is $\left(-9,\ \sqrt{2}\right)$.

Correct option: D

Final answer

$\left(-9,\ \sqrt{2}\right)$

Worked Solutions **Complex Numbers**

Q115 MCQ

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $2/5 - 3/4i$ using the relevant family rule.
2. The correct value is $\left(\frac{2}{5},\ -\frac{3}{4}\right)$.

Correct option: B

Final answer

$\left(\frac{2}{5},\ -\frac{3}{4}\right)$

Q116 MCQ

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{11}$ using the relevant family rule.
2. The correct value is $\left(-\sqrt{11},\ 0\right)$.

Correct option: B

Final answer

$\left(-\sqrt{11},\ 0\right)$

Worked Solutions **Complex Numbers**

Q117 *MCQ*

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{13}i$ using the relevant family rule.

2. The correct value is $\left(0,\ \sqrt{13}\right)$.

Correct option: C

Final answer

$\left(0,\ \sqrt{13}\right)$

Q118 *MCQ*

For $(2+i)x+(1-i)y=3+3i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.

2. Solving the real system gives $(2,\ -1)$.

Correct option: D

Final answer

$(2,\ -1)$

Worked Solutions **Complex Numbers**

Q119 MCQ

For $(1+2i)x+(3-i)y=5-4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1,\ 2)$.

Correct option: A

Final answer

$(-1,\ 2)$

Q120 MCQ

For $(3-2i)x+(1+i)y=6+i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1,\ 3)$.

Correct option: B

Final answer

$(1,\ 3)$

Worked Solutions **Complex Numbers**

Q121 *MCQ*

For $(2 - i)x + (4 + 2i)y = 4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-2,\ 1)$.

Correct option: B

Final answer

$(-2,\ 1)$

Q122 *MCQ*

For $(1 + i)x + (2 + 3i)y = 1$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(3,\ -1)$.

Correct option: C

Final answer

$(3,\ -1)$

Worked Solutions **Complex Numbers**

Q123 MCQ

For $(4+i)x+(1-2i)y=2+5i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1,\ -2)$.

Correct option: D

Final answer

$(1,\ -2)$

Q124 MCQ

For $(2+3i)x+(3+i)y=-8-5i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1,\ -2)$.

Correct option: D

Final answer

$(-1,\ -2)$

Worked Solutions **Complex Numbers**

Q125 *MCQ*

For $(1-i)x+(2-i)y=6-4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(2,\ 2)$.

Correct option: C

Final answer

$(2,\ 2)$

Q126 *MCQ*

For $(3+4i)-(1-2i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2=-1)$, reducing powers of i modulo 4.
2. The simplified value is $2+6i$.

Correct option: C

Final answer

$2+6i$

Worked Solutions **Complex Numbers**

Q127 MCQ

For $(-2+i)+(5-3i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2=-1)$, reducing powers of i modulo 4.
2. The simplified value is $3-2i$.

Correct option: C

Final answer

$3-2i$

Q128 MCQ

For $(1+2i)(3-i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2=-1)$, reducing powers of i modulo 4.
2. The simplified value is $5+5i$.

Correct option: D

Final answer

$5+5i$

Worked Solutions **Complex Numbers**

Q129 *MCQ*

For $(\sqrt{3}+i)(\sqrt{3}-i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 4.

Correct option: A

Final answer

4

Q130 *MCQ*

For $(2-i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2-i)^2$.

Correct option: C

Final answer

$(2-i)^2$

Worked Solutions **Complex Numbers**

Q131 MCQ

For $(1 + 3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-8 + 6i$.

Correct option: A

Final answer

$-8 + 6i$

Q132 MCQ

For i^{37}, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is i.

Correct option: B

Final answer

i

Worked Solutions **Complex Numbers**

Q133 MCQ

For $i^{22} + i^{9}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: B

Final answer

$-1 + i$

Q134 MCQ

For $i^{15} - i^{6}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $1 - i$.

Correct option: B

Final answer

$1 - i$

Worked Solutions **Complex Numbers**

Q135 MCQ

For $(4 - 2i) + (1 + 5i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2 = -1)$, reducing powers of i modulo 4.
2. The simplified value is $5 + 3i$.

Correct option: C

Final answer

$5 + 3i$

Q136 MCQ

For $(2 + i)(2 - i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2 = -1)$, reducing powers of i modulo 4.
2. The simplified value is 5.

Correct option: A

Final answer

5

Worked Solutions **Complex Numbers**

Q137 MCQ

For $(\sqrt{2}+2i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $\left(\sqrt{2}+2i\right)^2$.

Correct option: C

Final answer

$\left(\sqrt{2}+2i\right)^2$

Q138 MCQ

For $i^8 + i^{11} + i^{14}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-i$.

Correct option: C

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q139 MCQ

For $(3 - i)^3$, select the simplified standard form:

Solution

1. Expand and use $(i^2 = -1)$, reducing powers of i modulo 4.
2. The simplified value is $(3 - i)^3$.

Correct option: D

Final answer

$(3 - i)^3$

Q140 MCQ

For $(1 - 2i)(-2 + i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2 = -1)$, reducing powers of i modulo 4.
2. The simplified value is $5i$.

Correct option: D

Final answer

$5i$

Worked Solutions **Complex Numbers**

Q141 MCQ

For $i^5 + i^{10}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: D

Final answer

$-1 + i$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: D

Final answer

$2 + 6i$

Worked Solutions **Complex Numbers**

Q143 MCQ

For $(2-3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2-3i)^2$.

Correct option: C

Final answer

$(2-3i)^2$

Q144 MCQ

For the expression State the complex conjugate of $3-2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $3+2i$.

Correct option: C

Final answer

$3+2i$

Worked Solutions **Complex Numbers**

Q145 *MCQ*

For the expression State the complex conjugate of $-4+\sqrt{3}i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-4-\sqrt{3}i$.

Correct option: A

Final answer

$-4-\sqrt{3}i$

Q146 *MCQ*

For the expression State the complex conjugate of $7i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-7i$.

Correct option: C

Final answer

$-7i$

Worked Solutions **Complex Numbers**

Q147 MCQ

For the expression State the complex conjugate of $-\sqrt{5}$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\sqrt{5}$.

Correct option: C

Final answer

$-\sqrt{5}$

Q148 MCQ

For the expression State the complex conjugate of $1/2 - 3i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + 3i$.

Correct option: B

Final answer

$\frac{1}{2} + 3i$

Worked Solutions **Complex Numbers**

Q149 *MCQ*

For $(2+i)/(1-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2}+\frac{3i}{2}$.

Correct option: A

Final answer
$\frac{1}{2}+\frac{3i}{2}$

Q150 *MCQ*

For $(4-3i)/(2+i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $1-2i$.

Correct option: A

Final answer
$1-2i$

Worked Solutions **Complex Numbers**

Q151 MCQ

For $(1+2i)/(3-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{10} + \frac{7i}{10}$.

Correct option: B

Final answer

$\frac{1}{10} + \frac{7i}{10}$

Q152 MCQ

For $5i/(2-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $i\,(2+i)$.

Correct option: C

Final answer

$i\,(2+i)$

Worked Solutions **Complex Numbers**

Q153 *MCQ*

For $(3-i)/(1+3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D

Final answer

$-i$

Q154 *MCQ*

For $1/i$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q155 MCQ

For $(2+3i)/(2-3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13}+\frac{12i}{13}$.

Correct option: B

Final answer

$-\frac{5}{13}+\frac{12i}{13}$

Q156 MCQ

For $(\sqrt{2}+i)/(\sqrt{2}-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{2}+i}{\sqrt{2}-i}$.

Correct option: B

Final answer

$\frac{\sqrt{2}+i}{\sqrt{2}-i}$

Worked Solutions **Complex Numbers**

Q157 MCQ

For $(1-4i)/(3+2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13}-\frac{14i}{13}$.

Correct option: D

Final answer
$-\frac{5}{13}-\frac{14i}{13}$

Q158 MCQ

For $7i/(1-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{7}{2}+\frac{7i}{2}$.

Correct option: C

Final answer
$-\frac{7}{2}+\frac{7i}{2}$

Worked Solutions **Complex Numbers**

Q159 MCQ

For the expression State the complex conjugate of $-2 + 5i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-2 - 5i$.

Correct option: B

Final answer

$-2 - 5i$

Q160 MCQ

For the expression State the complex conjugate of $6 - \sqrt{7}i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $6 + \sqrt{7}i$.

Correct option: D

Final answer

$6 + \sqrt{7}i$

Worked Solutions **Complex Numbers**

Q161 *MCQ*

For the expression State the complex conjugate of $-i\sqrt{2}$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{2}i$.

Correct option: B

Final answer

$\sqrt{2}i$

Q162 *MCQ*

For the expression State the complex conjugate of 9:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is 9.

Correct option: C

Final answer

9

Worked Solutions **Complex Numbers**

Q163 MCQ

For the expression State the complex conjugate of $-3/4 + 2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{3}{4} - 2i$.

Correct option: A

Final answer

$-\frac{3}{4} - 2i$

Q164 MCQ

For $(5 + 2i)/(1 - 2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{5} + \frac{12i}{5}$.

Correct option: D

Final answer

$\frac{1}{5} + \frac{12i}{5}$

Worked Solutions **Complex Numbers**

Q165 MCQ

For $(2 - i)/(4 + i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{7}{17} - \frac{6i}{17}$.

Correct option: A

Final answer

$\frac{7}{17} - \frac{6i}{17}$

Q166 MCQ

For $3i/(2 + 3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{9}{13} + \frac{6i}{13}$.

Correct option: A

Final answer

$\frac{9}{13} + \frac{6i}{13}$

Worked Solutions **Complex Numbers**

Q167 MCQ

For $(\sqrt{3}-i)/(\sqrt{3}+i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\dfrac{\sqrt{3}-i}{\sqrt{3}+i}$.

Correct option: C

Final answer

$$\frac{\sqrt{3}-i}{\sqrt{3}+i}$$

Q168 MCQ

For $(1+i)/(2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\dfrac{1}{2}-\dfrac{i}{2}$.

Correct option: A

Final answer

$$\frac{1}{2}-\frac{i}{2}$$

Worked Solutions **Complex Numbers**

Q169 *MCQ*

For the expression State the complex conjugate of $4 - 7i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $4 + 7i$.

Correct option: D

Final answer

$4 + 7i$

Q170 *MCQ*

For the expression State the complex conjugate of $\sqrt{10} + 2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{10} - 2i$.

Correct option: A

Final answer

$\sqrt{10} - 2i$

Worked Solutions **Complex Numbers**

Q171 MCQ

For the expression State the complex conjugate of $-8i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $8i$.

Correct option: B

Final answer

$8i$

Q172 MCQ

For $(3+2i)/(3-2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{5}{13}+\frac{12i}{13}$.

Correct option: B

Final answer

$$\frac{5}{13}+\frac{12i}{13}$$

Worked Solutions **Complex Numbers**

Q173 **MCQ**

For $\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{5}$.

Correct option: B

Final answer

$i\sqrt{5}$

Q174 **MCQ**

For $\sqrt{-20}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{5}$.

Correct option: A

Final answer

$2i\sqrt{5}$

Worked Solutions **Complex Numbers**

Q175 MCQ

For $-\sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i\sqrt{2}$.

Correct option: A

Final answer

$-3i\sqrt{2}$

Q176 MCQ

For $squarerootsof-8$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{2}$.

Correct option: C

Final answer

$\pm 2i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 3.

Correct option: A

Final answer

3

Q178 MCQ

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -4.

Correct option: B

Final answer

-4

Worked Solutions **Complex Numbers**

Q179 MCQ

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: C

Final answer

$-2i$

Q180 MCQ

For $(2+\sqrt{-3})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $1+4i\sqrt{3}$.

Correct option: D

Final answer

$1+4i\sqrt{3}$

Worked Solutions **Complex Numbers**

Q181 MCQ

For $\sqrt{-72}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: D

Final answer

$6i\sqrt{2}$

Q182 MCQ

For $squarerootsof - 50$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 5i\sqrt{2}$.

Correct option: D

Final answer

$\pm 5i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q183 **MCQ**

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6.

Correct option: A

Final answer

-6

Q184 **MCQ**

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: D

Final answer

$-2i$

Worked Solutions **Complex Numbers**

Q185 MCQ

For $\sqrt{-32}+\sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: B

Final answer

$6i\sqrt{2}$

Q186 MCQ

For $\sqrt{-98}-\sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $4i\sqrt{2}$.

Correct option: B

Final answer

$4i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q187 MCQ

For $(\sqrt{5}+\sqrt{-2})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $3+2i\sqrt{10}$.

Correct option: A

Final answer

$3+2i\sqrt{10}$

Q188 MCQ

For $(3+\sqrt{-5})(3-\sqrt{-5})$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 14.

Correct option: D

Final answer

14

Worked Solutions **Complex Numbers**

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{7}$.

Correct option: C

Final answer
$i\sqrt{7}$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{7}$.

Correct option: A

Final answer
$2i\sqrt{7}$

Worked Solutions　　**Complex Numbers**

Q191 **MCQ**

For $squarerootsof - 12$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{3}$.

Correct option: A

Final answer

$\pm 2i\sqrt{3}$

Q192 **MCQ**

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i$.

Correct option: A

Final answer

$-3i$

Worked Solutions **Complex Numbers**

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6.

Correct option: D

Final answer

-6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-4i$.

Correct option: C

Final answer

$-4i$

Worked Solutions **Complex Numbers**

Q195 **MCQ**

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{2}$.

Correct option: C

Final answer

$5i\sqrt{2}$

Q196 **MCQ**

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-1 + 2i\sqrt{2}$.

Correct option: A

Final answer

$-1 + 2i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q197 MCQ

For $(4+\sqrt{-3})(4-\sqrt{-3})$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 19.

Correct option: B

Final answer

19

Q198 MCQ

For $\sqrt{-75}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{3}$.

Correct option: D

Final answer

$5i\sqrt{3}$

Worked Solutions **Complex Numbers**

Q199 MCQ

Solve for the real values x and y: $(1+3i)(x+i)=x(y+i)$**:**

Solution

1. Expand and equate real and imaginary parts.
2. The system gives $(x=-1/2)$ and $(y=7)$.

Correct option: B

Final answer

$\left(-\frac{1}{2},\ 7\right)$

Q200 MCQ

Given x is real, solve: $(4x-4i)(x^2+xi)=10x^2i^{32}$**:**

Solution

1. Use $(i^{32}=1)$ and $((x-i)(x+i)=x^2+1)$.
2. The real equation is $(4x(x^2+1)=10x^2)$, giving $x=0$, $1/2$, or 2.

Correct option: D

Final answer

$\left\{0,\ 2,\ \frac{1}{2}\right\}$

Worked Solutions **Complex Numbers**

Q201 MCQ

Simplify exactly: $(i(8-6i)/((1+2i)(3-i)))^{2027}$:

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The exact base simplifies to ((7+i)/5). Keep the 2027th power unexpanded.

Correct option: B

Final answer

$((7+i)/5)^{2027}$

Q202 MCQ

Solve over the complex numbers: $(3x-6)^2 = -8$:

Solution

1. Take square roots: ($3x - 6 = pm$ 2isqrt2).
2. Divide by 3 to obtain ($x = 2pm$
rac23isqrt2).

Correct option: B

Final answer

$x = 2 \pm (2/3)i\sqrt{2}$

Worked Solutions **Complex Numbers**

Quiz MCQ 1 **Concept Quiz · MCQ**

For 4-3i, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4 - 3i$ using the relevant family rule.
2. The correct value is $(4, \ -3)$.

Correct option: A

Final answer

$(4, \ -3)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For (1+2i)(2-i), select the simplified value:

Solution

1. Simplify $(1 + 2i)(2 - i)$ using the relevant family rule.
2. The correct value is $4 + 3i$.

Correct option: A

Final answer

$4 + 3i$

Worked Solutions **Complex Numbers**

Quiz MCQ 3 Concept Quiz · MCQ

For (3+i)/(1-i), select the value in the form a+bi:

Solution

1. Simplify $(3+i)/(1-i)$ using the relevant family rule.
2. The correct value is $1+2i$.

Correct option: D

Final answer

$1+2i$

Quiz MCQ 4 Concept Quiz · MCQ

For sqrt(-27), select the expression in terms of i:

Solution

1. Simplify $\sqrt{-27}$ using the relevant family rule.
2. The correct value is $3i\sqrt{3}$.

Correct option: A

Final answer

$3i\sqrt{3}$

Worked Solutions **Complex Numbers**

Quiz WRITTEN 5 **Concept Quiz · Written**

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $(5/2, -\sqrt{5}/2)$

Final answer

$(5/2, -\sqrt{5}/2)$

Quiz WRITTEN 6 **Concept Quiz · Written**

Solve for real x,y: $(2+i)x + (1-i)y = 3+i$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 4/3, y = 1/3$

Final answer

$x = 4/3, y = 1/3$

Worked Solutions **Complex Numbers**

Quiz WRITTEN 7 Concept Quiz · Written

Calculate $i^{18} + i^7 - i^3$ in standard form.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: -1

Final answer

-1

Quiz WRITTEN 8 Concept Quiz · Written

Solve over the complex numbers: $(x - 3)^2 = -12$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 3 \pm 2i\sqrt{3}$

Final answer

$x = 3 \pm 2i\sqrt{3}$

Answer key

Use the question number shown on each page.

Q001 $(3,\ -2)$

Q002 $(4/3, -\sqrt{5}/3)$

Q003 $(-8,\ 0)$

Q004 $(0,\ 2)$

Q005 $x = -1, y = 4$

Q006 $(-3,\ 5)$

Q007 $(-1/2, -\sqrt{3}/2)$

Q008 $(1,\ 0)$

Q009 $(0,\ 3)$

Q010 $x = 6, y = -2$

Q011 $x = -2, y = 3$

Q012 $(2,\ 5)$

Q013 $(-25,\ 13)$

Q014 $(2/3, -\sqrt{5}/3)$

Q015 $(7/2, -\sqrt{2}/2)$

Q016 $(-4,\ 0)$

Q017 $(1/4,\ 0)$

Q018 $(0, \sqrt{3})$

Q019 $(0,\ -1)$

Q020 $x = 3, y = 2$

Q021 $x = 3, y = -3$

Q022 $x = 2, y = -2$

Q023 $x = 2, y = 1$

Q024 $x = 1/2, y = 3/2$

Q025 $x = -1, y = 3$

Q026 $1 - 7i$

Q027 $3 - 4i$

Q028 -1

Q029 $1 + 8i$

Q030 $8 - i$

Q031 0

Q032 $4 - i$

Q033 $1 - 3i$

Q034 -2

Q035 $-1 + 4\sqrt{2}i$

Q036 $7 + 11i$

Q037 $1/15 - 13/14i$

Q038 $-3 - 4i$

Q039 -3

Q040 4

Q041 i

Q042 -1

Q043 $-i$

Q044 $x = 0$ or $x = 1$

Q045 $2 - 3i$

Q046 $7 + i\sqrt{11}$

Q047 6

Q048 $5i$

Q049 $3/5 + 4/5i$

Q050 $-1/2 - 3/2i$

Q051 $1 + 4i$

Q052 $\sqrt{2} - i\sqrt{10}$

Q053 2

Q054 $8i$

Q055 $1 - i$

Q056 $-1 - i$

Q057 $3 - 4i$

Q058 $9 - 8i$

Q059 $8 - i\sqrt{3}$

Q060 5

Q061 -15

Q062 $\sqrt{7}$

Q063 $-i$

Q064 $50i$

Q065 $-i\sqrt{6}$

Q066 i

Q067 $7/5 - 1/5i$

Q068 $2/3 + \sqrt{5}/3i$

Q069 $2 + i$

Q070 $3/5 + 4/5i$

Q071 $9/13 - 7/13i$

Q072 $-i$

Q073 $1/3 - 2/3i$

Q074 $((-2 - 144i)/85)^{2027}$

Q075 $\pm i\sqrt{6}$

Q076 $-6\sqrt{2}$

Q077 $-(2/3)i$

Q078 $i\sqrt{7}$

Q079 $3i\sqrt{3}$

Q080 $-2i\sqrt{3}$

Q081 $\pm 2i\sqrt{3}$

Q082 $-3\sqrt{2}$

Q083 $-3i$

Q084 $7 + 6\sqrt{2}i$

Q085 $i\sqrt{2}$

Q086 $3i$

Q087 $3i\sqrt{5}$

Q088 $-2i\sqrt{6}$

Q089 $-6i$

Q090 $\pm i\sqrt{2}$

Q091 $\pm 3i\sqrt{2}$

Q092 $\pm 4i$

Q093 $-2\sqrt{3}$

Q094 $-14\sqrt{5}$

Q095 $-2i$

Q096 $-(3/2)i$

Q097 $13i\sqrt{2}$

Q098 $6i\sqrt{2}$

Q099 $-2 + 2i\sqrt{15}$

Q100 7

Q101 $x = 4 \pm (\sqrt{10}/2)i$

Q102 C

Q103 D

Q104 D

Q105 B
Q106 D
Q107 A
Q108 B
Q109 C
Q110 B
Q111 D
Q112 D
Q113 C
Q114 D
Q115 B
Q116 B
Q117 C
Q118 D
Q119 A
Q120 B
Q121 B
Q122 C
Q123 D
Q124 D
Q125 C
Q126 C
Q127 C
Q128 D
Q129 A
Q130 C
Q131 A
Q132 B
Q133 B
Q134 B
Q135 C
Q136 A
Q137 C
Q138 C
Q139 D
Q140 D
Q141 D
Q142 D
Q143 C
Q144 C
Q145 A
Q146 C
Q147 C
Q148 B
Q149 A
Q150 A
Q151 B
Q152 C
Q153 D
Q154 D
Q155 B
Q156 B
Q157 D
Q158 C
Q159 B
Q160 D
Q161 B
Q162 C
Q163 A
Q164 D
Q165 A
Q166 A
Q167 C
Q168 A
Q169 D
Q170 A
Q171 B
Q172 B
Q173 B
Q174 A
Q175 A
Q176 C
Q177 A
Q178 B
Q179 C
Q180 D
Q181 D
Q182 D
Q183 A
Q184 D
Q185 B
Q186 B
Q187 A
Q188 D
Q189 C
Q190 A
Q191 A
Q192 A
Q193 D
Q194 C
Q195 C
Q196 A
Q197 B
Q198 D
Q199 B
Q200 D
Q201 B
Q202 B
Quiz MCQ 1 A
Quiz MCQ 2 A
Quiz MCQ 3 D
Quiz MCQ 4 A
Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Quiz WRITTEN 7 -1
Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a}, \ \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Quadratic Equations**

Q001 Written

Solve $x^2 + 8 = 0$.

Solution

1. $x^2 = -8$
2. $x = \pm\sqrt{-8} = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
3. Therefore the final answer is $x = \pm 2i\sqrt{2}$.

Final answer

$x = \pm 2i\sqrt{2}$

Q002 Written

Solve $x^2 + 4 = 0$.

Solution

1. $x^2 = -4$
2. $x = \pm 2i$
3. Therefore the final answer is $x = \pm 2i$.

Final answer

$x = \pm 2i$

Worked Solutions **Quadratic Equations**

Q003 **Written**

Solve $x^2 + 49 = 0$.

Solution

1. $x^2 = -49$
2. Therefore the final answer is $x = \pm 7i$.

Final answer

$x = \pm 7i$

Q004 **Written**

Solve $9x^2 - 4 = 0$.

Solution

1. $(3x - 2)(3x + 2) = 0$
2. Therefore the final answer is $x = \pm\frac{2}{3}$.

Final answer

$x = \pm\frac{2}{3}$

Worked Solutions **Quadratic Equations**

Q005 Written

Solve $x^2 - 12 = 0$.

Solution

1. $x = \pm\sqrt{12}$
2. Therefore the final answer is $x = \pm 2\sqrt{3}$.

Final answer

$x = \pm 2\sqrt{3}$

Q006 Written

Solve $4x^2 + 3 = 0$.

Solution

1. $x^2 = -3/4$
2. Therefore the final answer is $x = \pm\frac{i\sqrt{3}}{2}$.

Final answer

$x = \pm\frac{i\sqrt{3}}{2}$

Worked Solutions **Quadratic Equations**

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2i\sqrt{2}$.

Correct option: A

Final answer

$x = \pm 2i\sqrt{2}$

Q008 MCQ

Solve $x^2 + 4 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2i$.

Correct option: A

Final answer

$x = \pm 2i$

Worked Solutions **Quadratic Equations**

Q009 *MCQ*

Solve $x^2 + 49 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 7i$.

Correct option: C

Final answer

$x = \pm 7i$

Q010 *MCQ*

Solve $9x^2 - 4 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm\frac{2}{3}$.

Correct option: C

Final answer

$x = \pm\frac{2}{3}$

Worked Solutions **Quadratic Equations**

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2\sqrt{3}$.

Correct option: B

Final answer

$x = \pm 2\sqrt{3}$

Q012 MCQ

Solve $4x^2 + 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm \frac{i\sqrt{3}}{2}$.

Correct option: B

Final answer

$x = \pm \frac{i\sqrt{3}}{2}$

Worked Solutions **Quadratic Equations**

Q013 Written

Solve $(x-3)^2 = -8$.

Solution

1. $x - 3 = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
2. Therefore the final answer is $x = 3 \pm 2i\sqrt{2}$.

Final answer

$x = 3 \pm 2i\sqrt{2}$

Q014 Written

Solve $(x-2)^2 = 5$.

Solution

1. $x - 2 = \pm\sqrt{5}$
2. Therefore the final answer is $x = 2 \pm \sqrt{5}$.

Final answer

$x = 2 \pm \sqrt{5}$

Worked Solutions **Quadratic Equations**

Q015 Written

Solve $(x+6)^2 = -6$.

Solution

1. $x + 6 = \pm i\sqrt{6}$
2. Therefore the final answer is $x = -6 \pm i\sqrt{6}$.

Final answer

$x = -6 \pm i\sqrt{6}$

Q016 Written

Solve $(x+4)^2 = -18$.

Solution

1. $x + 4 = \pm i\sqrt{18} = \pm 3i\sqrt{2}$
2. Therefore the final answer is $x = -4 \pm 3i\sqrt{2}$.

Final answer

$x = -4 \pm 3i\sqrt{2}$

Worked Solutions **Quadratic Equations**

Q017 Written

Solve $(2x-3)^2 = 7$.

Solution

1. $2x - 3 = \pm\sqrt{7}$
2. Therefore the final answer is $x = \dfrac{3 \pm \sqrt{7}}{2}$.

Final answer

$x = \dfrac{3 \pm \sqrt{7}}{2}$

Q018 MCQ

Solve $(x-3)^2 = -8$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 3 \pm 2i\sqrt{2}$.

Correct option: A

Final answer

$x = 3 \pm 2i\sqrt{2}$

Worked Solutions **Quadratic Equations**

Q019 MCQ

Solve $(x-2)^2 = 5$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 2 \pm \sqrt{5}$.

Correct option: B

Final answer

$x = 2 \pm \sqrt{5}$

Q020 MCQ

Solve $(x+6)^2 = -6$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = -6 \pm i\sqrt{6}$.

Correct option: C

Final answer

$x = -6 \pm i\sqrt{6}$

Worked Solutions **Quadratic Equations**

Q021 *MCQ*

Solve $(x+4)^2 = -18$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = -4 \pm 3i\sqrt{2}$.

Correct option: C

Final answer

$x = -4 \pm 3i\sqrt{2}$

Q022 *MCQ*

Solve $(2x-3)^2 = 7$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.

2. The correct value is $x = \frac{3 \pm \sqrt{7}}{2}$.

Correct option: D

Final answer

$x = \frac{3 \pm \sqrt{7}}{2}$

Worked Solutions **Quadratic Equations**

Q023 **Written**

Solve $2x^2 - 7x - 9 = 0$.

Solution

1. $(x+1)(2x-9) = 0$
2. $x = -1 or x = 9/2$
3. Therefore the final answer is $x = -1,\ \dfrac{9}{2}$.

Final answer

$x = -1,\ \dfrac{9}{2}$

Q024 **Written**

Solve $2x^2 + 5x - 3 = 0$.

Solution

1. $(2x-1)(x+3) = 0$
2. Therefore the final answer is $x = \dfrac{1}{2},\ -3$.

Final answer

$x = \dfrac{1}{2},\ -3$

Worked Solutions **Quadratic Equations**

Q025 **Written**

Solve $2x^2 + 7x = 0$.

Solution

1. $x(2x + 7) = 0$
2. Therefore the final answer is $x = 0, -\frac{7}{2}$.

Final answer

$x = 0, -\frac{7}{2}$

Q026 **Written**

Solve $2x^2 + 5x - 12 = 0$.

Solution

1. $(2x - 3)(x + 4) = 0$
2. Therefore the final answer is $x = \frac{3}{2}, -4$.

Final answer

$x = \frac{3}{2}, -4$

Worked Solutions **Quadratic Equations**

Q027 Written

Solve $x^2 - 8x + 15 = 0$.

Solution

1. $(x-3)(x-5) = 0$
2. Therefore the final answer is $x = 3,\ 5$.

Final answer

$x = 3,\ 5$

Q028 MCQ

Solve $2x^2 - 7x - 9 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = -1,\ \frac{9}{2}$.

Correct option: A

Final answer

$x = -1,\ \frac{9}{2}$

Worked Solutions **Quadratic Equations**

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{1}{2}, -3$.

Correct option: D

Final answer

$x = \frac{1}{2}, -3$

Q030 MCQ

Solve $2x^2 + 7x = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 0, -\frac{7}{2}$.

Correct option: D

Final answer

$x = 0, -\frac{7}{2}$

Worked Solutions **Quadratic Equations**

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{3}{2}, -4$.

Correct option: A

Final answer

$x = \frac{3}{2}, -4$

Q032 MCQ

Solve $x^2 - 8x + 15 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 3, 5$.

Correct option: A

Final answer

$x = 3, 5$

Worked Solutions **Quadratic Equations**

Q033 Written

Solve $3x^2 - 4x - 1 = 0$.

Solution

1. $D = 16 + 12 = 28$
2. $x = (4 \pm 2\sqrt{7})/6 = (2 \pm \sqrt{7})/3$
3. Therefore the final answer is $x = \dfrac{2 \pm \sqrt{7}}{3}$.

Final answer

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Q034 MCQ

Solve $3x^2 - 4x - 1 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{2 \pm \sqrt{7}}{3}$.

Correct option: B

Final answer

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Worked Solutions **Quadratic Equations**

Q035 Written

Solve $2x^2 + 2\sqrt{6}\,x + 11 = 0.$

Solution

1. $D = (2\sqrt{6})^2 - 4(2)(11) = -64$
2. $x = (-2\sqrt{6} \pm \sqrt{(-64)})/4 = (-\sqrt{6} \pm 4i)/2$
3. Therefore the final answer is $x = \dfrac{-\sqrt{6} \pm 4i}{2}$.

Final answer

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Q036 Written

Solve $2x^2 + 2\sqrt{3}\,x + 5 = 0.$

Solution

1. $D = 12 - 40 = -28$
2. $x = (-2\sqrt{3} \pm i\sqrt{28})/4 = (-\sqrt{3} \pm i\sqrt{7})/2$
3. Therefore the final answer is $x = \dfrac{-\sqrt{3} \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Worked Solutions **Quadratic Equations**

Q037 Written

Solve $x^2 + x + 1 = 0$.

Solution

1. $D = 1 - 4 = -3$
2. $x = (-1 \pm i\sqrt{3})/2$
3. Therefore the final answer is $x = \dfrac{-1 \pm i\sqrt{3}}{2}$.

Final answer

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Q038 Written

Solve $3x^2 - 5x + 3 = 0$.

Solution

1. $D = 25 - 36 = -11$
2. $x = (5 \pm i\sqrt{11})/6$
3. Therefore the final answer is $x = \dfrac{5 \pm i\sqrt{11}}{6}$.

Final answer

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Worked Solutions **Quadratic Equations**

Q039 Written

Solve $x^2 - 2\sqrt{3}\,x + 5 = 0$**.**

Solution

1. $D = 12 - 20 = -8$
2. $x = (2\sqrt{3} \pm i\sqrt{8})/2 = \sqrt{3} \pm i\sqrt{2}$
3. Therefore the final answer is $x = \sqrt{3} \pm i\sqrt{2}$.

Final answer

$x = \sqrt{3} \pm i\sqrt{2}$

Q040 Written

Solve $3x^2 + 2\sqrt{2}\,x + 1 = 0$**.**

Solution

1. $D = 8 - 12 = -4$
2. $x = (-2\sqrt{2} \pm 2i)/6 = (-\sqrt{2} \pm i)/3$
3. Therefore the final answer is $x = \dfrac{-\sqrt{2} \pm i}{3}$.

Final answer

$x = \dfrac{-\sqrt{2} \pm i}{3}$

Worked Solutions **Quadratic Equations**

Q041 **Written**

Solve $x^2 - x + 2 = 0$.

Solution

1. $D = 1 - 8 = -7$
2. $x = (1 \pm i\sqrt{7})/2$
3. Therefore the final answer is $x = \dfrac{1 \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Q042 **Written**

Solve $3x^2 - \sqrt{5}\,x + 1 = 0$.

Solution

1. $D = 5 - 12 = -7$
2. $x = (\sqrt{5} \pm i\sqrt{7})/6$
3. Therefore the final answer is $x = \dfrac{\sqrt{5} \pm i\sqrt{7}}{6}$.

Final answer

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Worked Solutions **Quadratic Equations**

Q043 Written

Solve $6x^2 - 12x + 15 = 0$.

Solution

1. $D = 144 - 360 = -216$
2. $x = (12 \pm i6\sqrt{6})/12 = 1 \pm i\sqrt{6}/2$
3. Therefore the final answer is $x = 1 \pm \dfrac{i\sqrt{6}}{2}$.

Final answer

$x = 1 \pm \dfrac{i\sqrt{6}}{2}$

Q044 MCQ

Solve $2x^2 + 2\sqrt{6}\,x + 11 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-\sqrt{6} \pm 4i}{2}$.

Correct option: B

Final answer

$x = \dfrac{-\sqrt{6} \pm 4i}{2}$

Worked Solutions **Quadratic Equations**

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}\,x + 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-\sqrt{3} \pm i\sqrt{7}}{2}$.

Correct option: C

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-1 \pm i\sqrt{3}}{2}$.

Correct option: D

Final answer

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Worked Solutions **Quadratic Equations**

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{5 \pm i\sqrt{11}}{6}$.

Correct option: B

Final answer

$x = \dfrac{5 \pm i\sqrt{11}}{6}$

Q048 MCQ

Solve $x^2 - 2\sqrt{3}\,x + 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \sqrt{3} \pm i\sqrt{2}$.

Correct option: D

Final answer

$x = \sqrt{3} \pm i\sqrt{2}$

Worked Solutions **Quadratic Equations**

Q049 MCQ

Solve $3x^2 + 2\sqrt{2}\,x + 1 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-\sqrt{2} \pm i}{3}$.

Correct option: B

Final answer

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Q050 MCQ

Solve $x^2 - x + 2 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{1 \pm i\sqrt{7}}{2}$.

Correct option: C

Final answer

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Worked Solutions **Quadratic Equations**

Q051 MCQ

Solve $3x^2 - \sqrt{5}\,x + 1 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{\sqrt{5} \pm i\sqrt{7}}{6}$.

Correct option: C

Final answer

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 1 \pm \dfrac{i\sqrt{6}}{2}$.

Correct option: B

Final answer

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Worked Solutions **Quadratic Equations**

Q053 Written

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.

Solution

1. $D = 4^2 - 4(3)(7) = -68 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q054 Written

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.

Solution

1. $D = (-m)^2 - 4(m^2 - 3m - 9) = -3m^2 + 12m + 36$
2. $D >= 0 -> m^2 - 4m - 12 <= 0 -> (m + 2)(m - 6) <= 0$
3. Therefore the final answer is $-2 \leq m \leq 6$.

Final answer

$-2 \leq m \leq 6$

Q055 Written

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = k^2 - 4(3 - k) = (k - 2)(k + 6)$
2. $UseD > 0, D = 0, D < 0intervals$
3. Therefore the final answer is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

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Q056 Written

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.

Solution

1. $D = 25 - 48 = -23 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q057 **Written**

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.

Solution

1. $D = 49 + 32 = 81 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q058 **Written**

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.

Solution

1. $D = 100 - 100 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q059 Written

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.

Solution

1. $D = 25 - 12 = 13 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q060 Written

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.

Solution

1. $D = 144 - 144 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q061 **Written**

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.

Solution

1. $D = 64 - 80 = -16 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q062 **Written**

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.

Solution

1. $D = 16 - 16 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q063 Written

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Solution

1. $D = 9 - 20 = -11 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q064 Written

Determine the nature of the solutions of $x^2 - x - 2 = 0$.

Solution

1. $D = 1 + 8 = 9 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Worked Solutions **Quadratic Equations**

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-2 \leq m \leq 6$.

Correct option: D

Final answer

$-2 \leq m \leq 6$

Worked Solutions **Quadratic Equations**

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Correct option: C

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Worked Solutions **Quadratic Equations**

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: C

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions **Quadratic Equations**

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: A

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q071 *MCQ*

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: C

Final answer

Two distinct real solutions

Worked Solutions **Quadratic Equations**

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: C

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q073 *MCQ*

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: B

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: A

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q075 *MCQ*

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions **Quadratic Equations**

Q077 **Written**

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Solution

1. $D = 4m^2 - 4(m + 2)$
2. $D < 0- > m^2 - m - 2 < 0- > (m - 2)(m + 1) < 0$
3. Therefore the final answer is $-1 < m < 2$.

Final answer

$-1 < m < 2$

Q078 **Written**

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 - 4(2a - 3) = (a - 2)(a - 6)$
2. Classify by sign of D
3. Therefore the final answer is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Final answer

$a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions **Quadratic Equations**

Q079 Written

Find the values of a for which $x^2 - (a+1)x + 4 = 0$ has a repeated solution.

Solution

1. $D = (a+1)^2 - 16 = 0$
2. $a + 1 = \pm 4$
3. Therefore the final answer is $a = 3$ or $a = -5$.

Final answer

$a = 3$ or $a = -5$

Q080 Written

Find the range of m for which $x^2 + (m+6)x - 2m = 0$ has two distinct imaginary solutions.

Solution

1. $D = (m+6)^2 + 8m = m^2 + 20m + 36 = (m+2)(m+18)$
2. $D < 0$
3. Therefore the final answer is $-18 < m < -2$.

Final answer

$-18 < m < -2$

Worked Solutions **Quadratic Equations**

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Solution

1. $D = 4m^2 - 4(2m^2 - 3m) = 4m(3 - m)$
2. $D >= 0$
3. Therefore the final answer is $0 \leq m \leq 3$.

Final answer

$0 \leq m \leq 3$

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-1 < m < 2$.

Correct option: C

Final answer

$-1 < m < 2$

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Correct option: C

Final answer

$a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions **Quadratic Equations**

Q084 MCQ

Find the values of a for which $x^2 - (a+1)x + 4 = 0$ has a repeated solution Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a = 3$ or $a = -5$.

Correct option: A

Final answer

$a = 3$ or $a = -5$

Q085 MCQ

Find the range of m for which $x^2 + (m+6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-18 < m < -2$.

Correct option: D

Final answer

$-18 < m < -2$

Worked Solutions **Quadratic Equations**

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $0 \le m \le 3$.

Correct option: C

Final answer

$0 \le m \le 3$

Q087 Written

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.

Solution

1. $UseS = -b/a = 6/3 = 2andP = c/a = -5/3$
2. Therefore the final answer is $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$.

Final answer

$\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$

Worked Solutions **Quadratic Equations**

Q088 Written

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.

Solution

1. $S = -2/4 = -1/2; P = -7/4$
2. Therefore the final answer is $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$.

Final answer

$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$

Q089 Written

Find the sum and product of the roots of $x^2 + x + 2 = 0$.

Solution

1. $S = -1; P = 2$
2. Therefore the final answer is $S = -1, \quad P = 2$.

Final answer

$S = -1, \quad P = 2$

Worked Solutions **Quadratic Equations**

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.

Solution

1. $S = 7/3; P = -2$
2. Therefore the final answer is $S = \frac{7}{3}, \quad P = -2$.

Final answer
$S = \frac{7}{3}, \quad P = -2$

Q091 MCQ

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$.

Correct option: C

Final answer
$\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$

Worked Solutions **Quadratic Equations**

Q092 *MCQ*

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$.

Correct option: A

Final answer

$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$

Q093 *MCQ*

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = -1, \quad P = 2$.

Correct option: B

Final answer

$S = -1, \quad P = 2$

Worked Solutions **Quadratic Equations**

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = \frac{7}{3}, \quad P = -2$.

Correct option: A

Final answer

$S = \frac{7}{3}, \quad P = -2$

Q095 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 3, P = -3/2$
2. $S^2 - 2P = 9 + 3 = 12$
3. Therefore the final answer is 12.

Final answer

12

Worked Solutions **Quadratic Equations**

Q096 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = -3/2$
2. $S^3 - 3PS = 27 + 27/2 = 81/2$
3. Therefore the final answer is $\frac{81}{2}$.

Final answer

$\frac{81}{2}$

Q097 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3/2$
2. $S^2 - 2P = 4 - 3 = 1$
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions **Quadratic Equations**

Q098 **Written**

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3/2$
2. $S^3 - 3PS = 8 - 9 = -1$
3. Therefore the final answer is -1.

Final answer

-1

Q099 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3$
2. $S^2 - 2P = 4 - 6 = -2$
3. Therefore the final answer is -2.

Final answer

-2

Worked Solutions **Quadratic Equations**

Q100 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3$
2. $S^3 - 3PS = 8 - 18 = -10$
3. Therefore the final answer is -10.

Final answer

-10

Q101 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = -2, P = -1/3$
2. $S^2 - 2P = 4 + 2/3 = 14/3$
3. Therefore the final answer is $\frac{14}{3}$.

Final answer

$\frac{14}{3}$

Worked Solutions **Quadratic Equations**

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = -2, P = -1/3$
2. $S^3 - 3PS = -8 - 2 = -10$
3. Therefore the final answer is -10.

Final answer

-10

Q103 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 12.

Correct option: A

Final answer

12

Worked Solutions **Quadratic Equations**

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{81}{2}$.

Correct option: B

Final answer

$\frac{81}{2}$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 1.

Correct option: A

Final answer

1

Worked Solutions **Quadratic Equations**

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -1.

Correct option: B

Final answer

-1

Q107 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -2.

Correct option: D

Final answer

-2

Worked Solutions **Quadratic Equations**

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -10.

Correct option: B

Final answer

-10

Q109 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{14}{3}$.

Correct option: B

Final answer

$\frac{14}{3}$

Worked Solutions **Quadratic Equations**

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -10.

Correct option: D

Final answer

-10

Q111 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Solution

1. $2(\alpha + \beta)/(\alpha\beta) = 2(3)/(-3/2) = -4$
2. Therefore the final answer is -4.

Final answer

-4

Worked Solutions **Quadratic Equations**

Q112 **Written**

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Solution

1. $(\alpha^2 + \beta^2)/(\alpha\beta) = 1/(3/2) = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$\frac{2}{3}$

Q113 **Written**

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Solution

1. $S = 0; P = -3/5$
2. Therefore the final answer is $S = 0, \quad P = -\frac{3}{5}$.

Final answer

$S = 0, \quad P = -\frac{3}{5}$

Worked Solutions **Quadratic Equations**

Q114 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$\frac{2}{3}$

Q115 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = (-2)/(-1/3) = 6$
2. Therefore the final answer is 6.

Final answer

6

Worked Solutions **Quadratic Equations**

Q116 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -4.

Correct option: C

Final answer

-4

Q117 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D

Final answer

$\frac{2}{3}$

Worked Solutions **Quadratic Equations**

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = 0, \quad P = -\frac{3}{5}$.

Correct option: D

Final answer

$S = 0, \quad P = -\frac{3}{5}$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D

Final answer

$\frac{2}{3}$

Worked Solutions **Quadratic Equations**

Q120 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 6.

Correct option: D

Final answer

6

Q121 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Solution

1. $P + S + 1 = -3/2 + 3 + 1 = 5/2$
2. Therefore the final answer is $\frac{5}{2}$.

Final answer

$\frac{5}{2}$

Worked Solutions **Quadratic Equations**

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Solution

1. $P - S + 1 = 3/2 - 2 + 1 = 1/2$
2. Therefore the final answer is $\frac{1}{2}$.

Final answer

$\frac{1}{2}$

Q123 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Solution

1. $P + 2S + 4 = 3 + 4 + 4 = 11$
2. Therefore the final answer is 11.

Final answer

11

Worked Solutions **Quadratic Equations**

Q124 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Solution

1. $P - 2S + 4 = -1/3 + 4 + 4 = 23/3$
2. Therefore the final answer is $\frac{23}{3}$.

Final answer

$\frac{23}{3}$

Q125 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{5}{2}$.

Correct option: A

Final answer

$\frac{5}{2}$

Worked Solutions Quadratic Equations

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{1}{2}$.

Correct option: C

Final answer

$\frac{1}{2}$

Q127 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 11.

Correct option: D

Final answer

11

Worked Solutions **Quadratic Equations**

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{23}{3}$.

Correct option: D

Final answer

$\frac{23}{3}$

Q129 Written

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Solution

1. $Let roots \alpha, 3\alpha .4\alpha = 8 -> \alpha = 2.$
2. $k = \alpha(3\alpha) = 6$
3. Therefore the final answer is $k = 6, \quad x = 2, 6$.

Final answer

$k = 6, \quad x = 2, 6$

Worked Solutions **Quadratic Equations**

Q130 Written

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Solution

1. $Roots \alpha, 2\alpha; 3\alpha = -6 -> \alpha = -2; m = 2\alpha^2 = 8$
2. Therefore the final answer is $m = 8, \quad x = -2, -4$.

Final answer

$m = 8, \quad x = -2, -4$

Q131 Written

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Solution

1. $Roots \alpha, 2\alpha; 2\alpha^2 = 1/8 -> \alpha = \pm 1/4$.
2. $m = 8(3\alpha) = \pm 6$.
3. Therefore the final answer is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Final answer

$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Worked Solutions **Quadratic Equations**

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k = 6, \quad x = 2, 6$.

Correct option: A

Final answer

$k = 6, \quad x = 2, 6$

Q133 MCQ

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m = 8, \quad x = -2, -4$.

Correct option: A

Final answer

$m = 8, \quad x = -2, -4$

Worked Solutions **Quadratic Equations**

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Correct option: D

Final answer

$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Worked Solutions **Quadratic Equations**

Q135 **Written**

If the difference between the two roots of $x^2 - (m-1)x - 6 = 0$ is 5, find m and the roots.

Solution

1. $Let roots be \alpha and \alpha + 5.$
2. $\alpha(\alpha + 5) = -6 - > \alpha = -2 or - 3.$
3. $S = 2\alpha + 5 = m - 1 gives m = 2 or 0; obtain the paired roots.$
4. Therefore the final answer is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Final answer

$(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Q136 **Written**

If the difference between the roots of $x^2 - (m+1)x + 2 = 0$ is 1, find m and the roots.

Solution

1. $Let roots \alpha, \alpha + 1. \alpha(\alpha + 1) = 2 - > \alpha = 1 or - 2.$
2. $m + 1 = 2\alpha + 1 - > m = 2 or - 4.$
3. Therefore the final answer is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Final answer

$(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Worked Solutions   **Quadratic Equations**

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Solution

1. $Roots\alpha, \alpha + 4; 2\alpha + 4 = -6- > \alpha = -5; m = \alpha(\alpha + 4) = 5$
2. Therefore the final answer is $m = 5, \quad x = -5, -1$.

Final answer

$m = 5, \quad x = -5, -1$

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Solution

1. $Roots\alpha, \alpha + 1; \alpha(\alpha + 1) = 3/4- > \alpha = 1/2or - 3/2.$
2. $-m/4 = 2\alpha + 1- > m = -8or8.$
3. Therefore the final answer is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Final answer

$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Worked Solutions **Quadratic Equations**

Q139 MCQ

If the difference between the two roots of $x^2 - (m-1)x - 6 = 0$ is 5, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Correct option: D

Final answer

$(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Worked Solutions **Quadratic Equations**

Q140 MCQ

If the difference between the roots of $x^2 - (m+1)x + 2 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Correct option: A

Final answer

$(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Worked Solutions **Quadratic Equations**

Q141 **MCQ**

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m = 5, \quad x = -5, -1$.

Correct option: B

Final answer

$m = 5, \quad x = -5, -1$

Worked Solutions **Quadratic Equations**

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Correct option: B

Final answer

$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Worked Solutions **Quadratic Equations**

Q143 Written

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots.

Solution

1. $Letroots 2t, 3t; 5t = 2 -> t = 2/5; m = 6t^2 = 24/25$
2. Therefore the final answer is $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$.

Final answer

$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

Q144 MCQ

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$.

Correct option: A

Final answer

$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

Worked Solutions **Quadratic Equations**

Q145 Written

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Solution

1. $Roots\alpha, \alpha^2; \alpha + \alpha^2 = 6- > \alpha = 2or - 3.$
2. $k = \alpha^3 = 8or - 27.$
3. Therefore the final answer is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Final answer

$(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Worked Solutions **Quadratic Equations**

Q146 *MCQ*

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Correct option: B

Final answer

$(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Worked Solutions **Quadratic Equations**

Q147 Written

Factorise $2x^2 - 3x + 2$ over the complex numbers.

Solution

1. $D = 9 - 16 = -7; roots(3 \pm i\sqrt{7})/4$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$.

Final answer

$2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$

Q148 Written

Factorise $x^2 + 2x + 5$ over the complex numbers.

Solution

1. $Roots - 1 \pm 2i$
2. Therefore the final answer is $(x + 1 - 2i)(x + 1 + 2i)$.

Final answer

$(x + 1 - 2i)(x + 1 + 2i)$

Worked Solutions **Quadratic Equations**

Q149 Written

Factorise $2x^2 - 6x + 1$ over the complex numbers.

Solution

1. $D = 28; roots(3 \pm \sqrt{7})/2$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$.

Final answer

$$2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$$

Q150 Written

Factorise $x^2 + 4x + 5$ over the complex numbers.

Solution

1. $D = 16 - 20 = -4; roots - 2 \pm i$
2. Therefore the final answer is $(x + 2 - i)(x + 2 + i)$.

Final answer

$$(x + 2 - i)(x + 2 + i)$$

Q151 Written

Factorise $x^2 + 4x + 2$ over the complex numbers.

Solution

1. $Roots - 2 \pm \sqrt{2}$
2. Therefore the final answer is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Final answer

$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

Q152 Written

Factorise $x^2 - 4x + 7$ over the complex numbers.

Solution

1. $D = 16 - 28 = -12; roots2 \pm i\sqrt{3}$
2. Therefore the final answer is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Final answer

$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

Worked Solutions **Quadratic Equations**

Q153 Written

Factorise $2x^2 - x + 1$ over the complex numbers.

Solution

1. $D = 1 - 8 = -7; roots(1 \pm i\sqrt{7})/4$
2. Therefore the final answer is $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$.

Final answer

$2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$

Q154 Written

Factorise $3x^2 - 5x + 3$ over the complex numbers.

Solution

1. $D = 25 - 36 = -11; roots(5 \pm i\sqrt{11})/6$
2. Therefore the final answer is $3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$.

Final answer

$3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$

Worked Solutions **Quadratic Equations**

Q155 **Written**

Factorise $2x^2 - 2x - 1$ over the complex numbers.

Solution

1. $D = 4 + 8 = 12; roots (1 \pm \sqrt{3})/2$
2. Therefore the final answer is $2\left(x - \dfrac{1+\sqrt{3}}{2}\right)\left(x - \dfrac{1-\sqrt{3}}{2}\right)$.

Final answer

$$2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$$

Worked Solutions **Quadratic Equations**

Q156 *MCQ*

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Correct option: C

Final answer

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 1 - 2i)(x + 1 + 2i)$.

Correct option: A

Final answer

$(x + 1 - 2i)(x + 1 + 2i)$

Worked Solutions **Quadratic Equations**

Q158 *MCQ*

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$.

Correct option: D

Final answer

$$2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$$

Worked Solutions **Quadratic Equations**

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 2 - i)(x + 2 + i)$.

Correct option: D

Final answer

$(x + 2 - i)(x + 2 + i)$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Correct option: B

Final answer

$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

Worked Solutions **Quadratic Equations**

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Correct option: A

Final answer

$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

Worked Solutions **Quadratic Equations**

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.

Correct option: C

Final answer

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Worked Solutions **Quadratic Equations**

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $3\left(x - \dfrac{5+i\sqrt{11}}{6}\right)\left(x - \dfrac{5-i\sqrt{11}}{6}\right)$.

Correct option: A

Final answer

$$3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$$

Worked Solutions **Quadratic Equations**

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \dfrac{1+\sqrt{3}}{2}\right)\left(x - \dfrac{1-\sqrt{3}}{2}\right)$.

Correct option: B

Final answer

$$2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$$

Worked Solutions **Quadratic Equations**

Q165 **Written**

Factorise $x^4 + 2x^2 - 3$ over the complex numbers.

Solution

1. $x^4 + 2x^2 - 3 = (x^2 - 1)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$.

Final answer

$(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$

Q166 **Written**

Factorise $x^4 - x^2 - 6$ over the complex numbers.

Solution

1. $Let y = x^2 : y^2 - y - 6 = (y-3)(y+2)$
2. $Factor x^2 - 3 and x^2 + 2$
3. Therefore the final answer is $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$.

Final answer

$(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$

Worked Solutions **Quadratic Equations**

Q167 Written

Factorise $x^4 - 9$ over the complex numbers.

Solution

1. $x^4 - 9 = (x^2 - 3)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$

Q168 Written

Factorise $x^4 - 3x^2 + 2$ over the complex numbers.

Solution

1. $Let y = x^2 : y^2 - 3y + 2 = (y - 1)(y - 2)$
2. $Factor x^2 - 1 and x^2 - 2$
3. Therefore the final answer is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Final answer

$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$

Worked Solutions **Quadratic Equations**

Q169 *MCQ*

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$.

Correct option: B

Final answer

$(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$

Q170 *MCQ*

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$.

Correct option: B

Final answer

$(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$

Worked Solutions **Quadratic Equations**

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: D

Final answer

$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Correct option: C

Final answer

$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$

Worked Solutions **Quadratic Equations**

Q173 **Written**

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Solution

1. $S = 1, P = 1/2$
2. $x^2 - x + 1/2 = 0; multiplyby2$
3. Therefore the final answer is $2x^2 - 2x + 1 = 0$.

Final answer

$2x^2 - 2x + 1 = 0$

Q174 **Written**

Find two numbers whose sum is -1 and product is 2.

Solution

1. $Theyarerootsof x^2 + x + 2 = 0$
2. $D = -7$
3. Therefore the final answer is $\dfrac{-1 + i\sqrt{7}}{2}, \dfrac{-1 - i\sqrt{7}}{2}$.

Final answer

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Worked Solutions **Quadratic Equations**

Q175 Written

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Solution

1. $S = 6, P = 13$
2. $x^2 - Sx + P = 0$
3. Therefore the final answer is $x^2 - 6x + 13 = 0$.

Final answer

$x^2 - 6x + 13 = 0$

Q176 Written

Find two numbers whose sum and product are both 1.

Solution

1. $Roots of x^2 - x + 1 = 0$
2. $D = -3$
3. Therefore the final answer is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.

Final answer

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Worked Solutions **Quadratic Equations**

Q177 Written

Create a quadratic equation with roots -1 and 2.

Solution

1. $S = 1, P = -2$
2. Therefore the final answer is $x^2 - x - 2 = 0$.

Final answer

$x^2 - x - 2 = 0$

Q178 Written

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.

Solution

1. $S = 2, P = 17$
2. Therefore the final answer is $x^2 - 2x + 17 = 0$.

Final answer

$x^2 - 2x + 17 = 0$

Worked Solutions **Quadratic Equations**

Q179 Written

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Solution

1. $S = 10/3, P = 2$
2. $x^2 - (10/3)x + 2 = 0; multiplyby3$
3. Therefore the final answer is $3x^2 - 10x + 6 = 0$.

Final answer

$3x^2 - 10x + 6 = 0$

Q180 Written

Find two numbers whose sum is 2 and product is 5.

Solution

1. $Rootsof x^2 - 2x + 5 = 0$
2. $D = -16$
3. Therefore the final answer is $1 + 2i,\ 1 - 2i$.

Final answer

$1 + 2i,\ 1 - 2i$

Worked Solutions **Quadratic Equations**

Q181 Written

Find two numbers whose sum is 4 and product is 2.

Solution

1. $Rootsof\,x^2 - 4x + 2 = 0$
2. Therefore the final answer is $2 + \sqrt{2},\ 2 - \sqrt{2}$.

Final answer

$2 + \sqrt{2},\ 2 - \sqrt{2}$

Q182 MCQ

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2x^2 - 2x + 1 = 0$.

Correct option: B

Final answer

$2x^2 - 2x + 1 = 0$

Worked Solutions **Quadratic Equations**

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\dfrac{-1+i\sqrt{7}}{2}, \dfrac{-1-i\sqrt{7}}{2}$.

Correct option: B

Final answer

$\dfrac{-1+i\sqrt{7}}{2}, \dfrac{-1-i\sqrt{7}}{2}$

Q184 MCQ

Create a quadratic equation with roots $3+2i$ and $3-2i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 6x + 13 = 0$.

Correct option: B

Final answer

$x^2 - 6x + 13 = 0$

Worked Solutions **Quadratic Equations**

Q185 *MCQ*

Find two numbers whose sum and product are both 1 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$.

Correct option: A

Final answer

$\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

Q186 *MCQ*

Create a quadratic equation with roots -1 and 2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - x - 2 = 0$.

Correct option: A

Final answer

$x^2 - x - 2 = 0$

Worked Solutions **Quadratic Equations**

Q187 MCQ

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 2x + 17 = 0$.

Correct option: D

Final answer

$x^2 - 2x + 17 = 0$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $3x^2 - 10x + 6 = 0$.

Correct option: A

Final answer

$3x^2 - 10x + 6 = 0$

Worked Solutions **Quadratic Equations**

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $1 + 2i,\ 1 - 2i$.

Correct option: B

Final answer

$1 + 2i,\ 1 - 2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2 + \sqrt{2},\ 2 - \sqrt{2}$.

Correct option: B

Final answer

$2 + \sqrt{2},\ 2 - \sqrt{2}$

Worked Solutions **Quadratic Equations**

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Solution

1. $S = -2, P = -4$
2. $Newsum = S + 4 = 2; newproduct = P + 2S + 4 = -4$
3. $Equationx^2 - 2x - 4 = 0$
4. Therefore the final answer is $x^2 - 2x - 4 = 0$.

Final answer

$x^2 - 2x - 4 = 0$

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 2x - 4 = 0$.

Correct option: C

Final answer

$x^2 - 2x - 4 = 0$

Worked Solutions **Quadratic Equations**

Q193 **Written**

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2.

Solution

1. $S = -1, P = -3$
2. $NewsumS^2 - 2P = 1 + 6 = 7; productP^2 = 9$
3. Therefore the final answer is $x^2 - 7x + 9 = 0$.

Final answer

$x^2 - 7x + 9 = 0$

Q194 **Written**

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2.

Solution

1. $S = 2, P = 7$
2. $Newsum = S^2 - 2P = -10; product = P^2 = 49$
3. Therefore the final answer is $x^2 + 10x + 49 = 0$.

Final answer

$x^2 + 10x + 49 = 0$

Worked Solutions **Quadratic Equations**

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 7x + 9 = 0$.

Correct option: D

Final answer

$x^2 - 7x + 9 = 0$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 + 10x + 49 = 0$.

Correct option: C

Final answer

$x^2 + 10x + 49 = 0$

Worked Solutions **Quadratic Equations**

Q197 Written

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Solution

1. $S = -3/2, P = 2$
2. $Newsum = S/P = -3/4; product = 1/P = 1/2$
3. $x^2 + (3/4)x + 1/2 = 0; multiplyby4$
4. Therefore the final answer is $4x^2 + 3x + 2 = 0$.

Final answer

$4x^2 + 3x + 2 = 0$

Q198 MCQ

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $4x^2 + 3x + 2 = 0$.

Correct option: A

Final answer

$4x^2 + 3x + 2 = 0$

Worked Solutions **Quadratic Equations**

Q199 Written

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Solution

1. $S = -2, P = 3$
2. $Newsum = 4S = -8$
3. $Newproduct = 3(\alpha^2 + \beta^2) + 10P = 3(S^2 - 2P) + 10P = 24$
4. Therefore the final answer is $x^2 + 8x + 24 = 0$.

Final answer

$x^2 + 8x + 24 = 0$

Q200 MCQ

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 + 8x + 24 = 0$.

Correct option: D

Final answer

$x^2 + 8x + 24 = 0$

Worked Solutions **Quadratic Equations**

Q201 **Written**

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m.

Solution

1. $D > 0 -> (m-2)(m+1) > 0 -> m < -1 or m > 2$
2. $S = -2m > 0 -> m < 0$
3. $P = m + 2 > 0 -> m > -2$
4. $Intersection : -2 < m < -1$
5. Therefore the final answer is $-2 < m < -1$.

Final answer

$-2 < m < -1$

Worked Solutions **Quadratic Equations**

Q202 **Written**

If $x^2 + 2(m-1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m.

Solution

1. $D = 4(4-m)(m+1) > 0- > -1 < m < 4$
2. $S = 2(1-m) > 0- > m < 1$
3. $P = (2m+1)(m-3) > 0- > m < -1/2 or m > 3$
4. $Intersection : -1 < m < -1/2$
5. Therefore the final answer is $-1 < m < -\frac{1}{2}$.

Final answer

$-1 < m < -\frac{1}{2}$

Worked Solutions **Quadratic Equations**

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-2 < m < -1$.

Correct option: A

Final answer

$-2 < m < -1$

Q204 MCQ

If $x^2 + 2(m-1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-1 < m < -\frac{1}{2}$.

Correct option: B

Final answer

$-1 < m < -\frac{1}{2}$

Worked Solutions **Quadratic Equations**

Q205 Written

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m.

Solution

1. $D > 0 : 4m^2 - 4(m+6) > 0 -> (m-3)(m+2) > 0$
2. $S = 2m < 0 -> m < 0$
3. $P = m + 6 > 0 -> m > -6$
4. $Intersection : -6 < m < -2$
5. Therefore the final answer is $-6 < m < -2$.

Final answer

$-6 < m < -2$

Worked Solutions **Quadratic Equations**

Q206 Written

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k.

Solution

1. $D = (k - 10)(k + 2) > 0 - > k < -2 or k > 10$
2. $S = -k < 0 - > k > 0$
3. $P = 2k + 5 > 0 - > k > -5/2$
4. $Intersection k > 10$
5. Therefore the final answer is $k > 10$.

Final answer

$k > 10$

Worked Solutions **Quadratic Equations**

Q207 MCQ

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-6 < m < -2$.

Correct option: C

Final answer

$-6 < m < -2$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k > 10$.

Correct option: B

Final answer

$k > 10$

Worked Solutions **Quadratic Equations**

Q209 Written

If $x^2 - (m-3)x - m - 4 = 0$ has roots of different signs, find the range of m.

Solution

1. $Opposite signs iff P = -m - 4 < 0$
2. Therefore the final answer is $m > -4$.

Final answer

$m > -4$

Q210 Written

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k.

Solution

1. $P = -k - 3 < 0$
2. Therefore the final answer is $k > -3$.

Final answer

$k > -3$

Worked Solutions **Quadratic Equations**

Q211 MCQ

If $x^2 - (m-3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m > -4$.

Correct option: A

Final answer

$m > -4$

Q212 MCQ

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k > -3$.

Correct option: D

Final answer

$k > -3$

Worked Solutions **Quadratic Equations**

Q213 **Challenge**

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 + 4a = a(a+4)$
2. Classify sign intervals at -4 and 0
3. Therefore the final answer is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions **Quadratic Equations**

Q214 **Challenge**

For $x^2 - (k+2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = (k+2)^2 - 4k^2 = -3k^2 + 4k + 4 = -3(k-2)(k+2/3)$
2. Classify D sign
3. Therefore the final answer is $-\frac{2}{3} < k < 2:$ distinct real; $k = -\frac{2}{3}, 2:$ repeated; $k < -\frac{2}{3}$ or $k > 2:$ distinct imaginary.

Final answer

$-\frac{2}{3} < k < 2:$ distinct real; $k = -\frac{2}{3}, 2:$ repeated; $k < -\frac{2}{3}$ or $k > 2:$ distinct imaginary

Worked Solutions **Quadratic Equations**

Q215 Challenge

For $x^2 - 2(a-1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = 4(a-1)^2 - 4(a^2 - 3a + 4) = 4(a-3)$
2. Classify by sign of a-3
3. Therefore the final answer is $a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary.

Final answer

$a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary

Worked Solutions **Quadratic Equations**

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Solution

1. $\alpha\beta(\alpha + \beta) = PS = (3/2)(3) = 9/2$
2. Therefore the final answer is $\frac{9}{2}$.

Final answer

$\frac{9}{2}$

Worked Solutions **Quadratic Equations**

Q217 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Solution

1. $S^2 - 4P = 9 - 6 = 3$
2. Therefore the final answer is 3.

Final answer

3

Worked Solutions **Quadratic Equations**

Q218 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = 3/2$
2. $S^3 - 3PS = 27 - 27/2 = 27/2$
3. Therefore the final answer is $\frac{27}{2}$.

Final answer

$\frac{27}{2}$

Worked Solutions **Quadratic Equations**

Q219 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Solution

1. $(\alpha^3 + \beta^3)/(\alpha\beta) = (27/2)/(3/2) = 9$
2. Therefore the final answer is 9.

Final answer

9

Worked Solutions **Quadratic Equations**

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Solution

1. $Subtractequations : (a - b)(x - c) = 0, socommonrootr = c.$
2. $Substitutec : c(c + a + b) = 0; c \neq 0 - > a + b + c = 0.$
3. $Theotherrootsarebc/c = bandca/c = a.Theirequationisx^2 - (a + b)x + ab = x^2 + cx + ab.$
4. Therefore the final answer is Common root c; other roots a and b..

Final answer

Common root c; other roots a and b.

Worked Solutions **Quadratic Equations**

Q221 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Solution

1. $Let y = x^2 : 6y^2 - 7y - 3 = (2y - 3)(3y + 1)$
2. $x^2 = 3/2 or - 1/3; hence x = \pm\sqrt{6}/2 or \pm i\sqrt{3}/3$
3. $For real displacement x, only \pm\sqrt{6}/2 are admissible$
4. Therefore the final answer is $x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$.

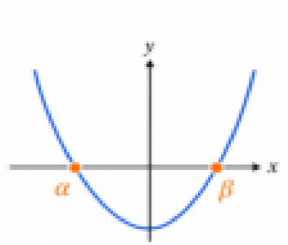
Final answer

$x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$

Worked Solutions **Quadratic Equations**

Q222 **Challenge**

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Solution

1. $Newsum = S/P; newproduct = 1/P$
2. $Equationx^2 - (S/P)x + 1/P = 0; multiplybyP$
3. Therefore the final answer is For $P = \alpha\beta \neq 0 : \quad Px^2 - Sx + 1 = 0.$

Final answer

For $P = \alpha\beta \neq 0 : \quad Px^2 - Sx + 1 = 0$

Worked Solutions **Quadratic Equations**

Q223 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Solution

1. $D = (m-5)(m+3) > 0- > m < -3 or m > 5$
2. $S = -(m+1) > 0- > m < -1$
3. $P = m + 4 > 0- > m > -4$
4. Intersection (-4,-3)
5. Therefore the final answer is $-4 < m < -3$.

Final answer

$-4 < m < -3$

Worked Solutions **Quadratic Equations**

Q224 **Challenge**

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Solution

1. $D > 0 -> m < -3 or m > 5$
2. $S < 0 -> m > -1$
3. $P > 0 -> m > -4$
4. $Intersection m > 5$
5. Therefore the final answer is $m > 5$.

Final answer

$m > 5$

Worked Solutions **Quadratic Equations**

Q225 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Solution

1. $Oppositesignsiff P = m + 4 < 0$
2. Therefore the final answer is $m < -4$.

Final answer

$m < -4$

Worked Solutions **Quadratic Equations**

Q226 Challenge

A model has $R = 3x^2 + 2\sqrt{3}\,x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Solution

1. $Set R = 0. D = 12 - 24 = -12.$
2. $x = (-2\sqrt{3} \pm i2\sqrt{3})/6 = (-\sqrt{3} \pm i\sqrt{3})/3.$
3. Because x is real, the complex pair is not admissible; no real equilibrium exists.
4. Therefore the final answer is $x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Final answer

$x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Worked Solutions **Quadratic Equations**

Q227 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Correct option: B

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions **Quadratic Equations**

Q228 **Challenge**

For $x^2 - (k+2)x + k^2 = 0$, determine the nature of the solutions as k varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-\frac{2}{3} < k < 2:$ distinct real; $k = -\frac{2}{3}, 2:$ repeated; $k < -\frac{2}{3}$ or $k > 2:$ distinct imaginary.

Correct option: B

Final answer

$-\frac{2}{3} < k < 2:$ distinct real; $k = -\frac{2}{3}, 2:$ repeated; $k < -\frac{2}{3}$ or $k > 2:$ distinct imaginary

Worked Solutions **Quadratic Equations**

Q229 **Challenge**

For $x^2 - 2(a-1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary.

Correct option: B

Final answer

$a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary

Worked Solutions **Quadratic Equations**

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{9}{2}$.

Correct option: B

Final answer

$\frac{9}{2}$

Worked Solutions **Quadratic Equations**

Q231 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 3.

Correct option: D

Final answer

3

Worked Solutions **Quadratic Equations**

Q232 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{27}{2}$.

Correct option: B

Final answer

$\frac{27}{2}$

Worked Solutions **Quadratic Equations**

Q233 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 9.

Correct option: A

Final answer

9

Worked Solutions **Quadratic Equations**

Q234 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Common root c; other roots a and b..

Correct option: C

Final answer

Common root c; other roots a and b.

Worked Solutions **Quadratic Equations**

Q235 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$.

Correct option: A

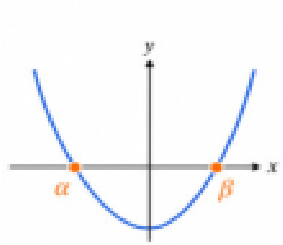
Final answer

$x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$

Worked Solutions **Quadratic Equations**

Q236 **Challenge**

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ Select the correct answer:



Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is For $P = \alpha\beta \neq 0: \quad Px^2 - Sx + 1 = 0$.

Correct option: A

Final answer

For $P = \alpha\beta \neq 0: \quad Px^2 - Sx + 1 = 0$

Worked Solutions **Quadratic Equations**

Q237 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-4 < m < -3$.

Correct option: C

Final answer

$-4 < m < -3$

Worked Solutions **Quadratic Equations**

Q238 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m > 5$.

Correct option: D

Final answer

$m > 5$

Worked Solutions **Quadratic Equations**

Q239 **Challenge**

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two roots of opposite signs Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m < -4$.

Correct option: D

Final answer

$m < -4$

Worked Solutions **Quadratic Equations**

Q240 Challenge

A model has $R = 3x^2 + 2\sqrt{3}\,x + 2$. Determine the values of x for which $R = 0$, and interpret the result Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Correct option: D

Final answer

$x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Worked Solutions **Quadratic Equations**

Quiz MCQ 1 Concept Quiz · MCQ

Solve $x^2 - 5x + 6 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2, 3$.

Correct option: B

Final answer

$2, 3$

Worked Solutions **Quadratic Equations**

Quiz MCQ 2 Concept Quiz · MCQ

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct non-real complex roots.

Correct option: A

Final answer

Two distinct non-real complex roots

Worked Solutions **Quadratic Equations**

Quiz MCQ 3 **Concept Quiz · MCQ**

If alpha,beta are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 7.

Correct option: C

Final answer

7

Quiz MCQ 4 **Concept Quiz · MCQ**

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 4x + 5 = 0$.

Correct option: A

Final answer

$x^2 - 4x + 5 = 0$

Worked Solutions **Quadratic Equations**

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $2x^2 + 3x - 2 = 0$.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $x = \frac{1}{2}, -2$

Final answer

$x = \frac{1}{2}, -2$

Quiz WRITTEN 6 **Concept Quiz · Written**

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $-1 < m < 2$

Final answer

$-1 < m < 2$

Worked Solutions **Quadratic Equations**

Quiz WRITTEN 7 **Concept Quiz · Written**

If alpha,beta are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: 28

Final answer

28

Quiz WRITTEN 8 **Concept Quiz · Written**

If alpha,beta are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $x^2 + 6x + 25 = 0$

Final answer

$x^2 + 6x + 25 = 0$

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$
Q002 $x = \pm 2i$
Q003 $x = \pm 7i$
Q004 $x = \pm\frac{2}{3}$
Q005 $x = \pm 2\sqrt{3}$
Q006 $x = \pm\frac{i\sqrt{3}}{2}$
Q007 A
Q008 A
Q009 C
Q010 C
Q011 B
Q012 B
Q013 $x = 3 \pm 2i\sqrt{2}$
Q014 $x = 2 \pm \sqrt{5}$
Q015 $x = -6 \pm i\sqrt{6}$
Q016 $x = -4 \pm 3i\sqrt{2}$
Q017 $x = \frac{3 \pm \sqrt{7}}{2}$
Q018 A
Q019 B
Q020 C
Q021 C
Q022 D
Q023 $x = -1,\ \frac{9}{2}$
Q024 $x = \frac{1}{2},\ -3$
Q025 $x = 0,\ -\frac{7}{2}$
Q026 $x = \frac{3}{2},\ -4$
Q027 $x = 3,\ 5$
Q028 A
Q029 D
Q030 D
Q031 A
Q032 A
Q033 $x = \frac{2 \pm \sqrt{7}}{3}$
Q034 B
Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$
Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$
Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$
Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$
Q039 $x = \sqrt{3} \pm i\sqrt{2}$
Q040 $x = \frac{-\sqrt{2} \pm i}{3}$
Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$
Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$
Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$
Q044 B
Q045 C
Q046 D
Q047 B
Q048 D
Q049 B
Q050 C
Q051 C
Q052 B
Q053 Two distinct imaginary (non-real complex) solutions
Q054 $-2 \le m \le 6$
Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary
Q056 Two distinct imaginary (non-real complex) solutions
Q057 Two distinct real solutions
Q058 One repeated real solution
Q059 Two distinct real solutions
Q060 One repeated real solution
Q061 Two distinct imaginary (non-real complex) solutions
Q062 One repeated real solution
Q063 Two distinct imaginary (non-real complex) solutions
Q064 Two distinct real solutions
Q065 A
Q066 D
Q067 C
Q068 C
Q069 A
Q070 A
Q071 C
Q072 C
Q073 B
Q074 A
Q075 A
Q076 A
Q077 $-1 < m < 2$
Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary
Q079 $a = 3$ or $a = -5$
Q080 $-18 < m < -2$
Q081 $0 \le m \le 3$
Q082 C
Q083 C
Q084 A
Q085 D
Q086 C
Q087 $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$
Q088 $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$
Q089 $S = -1, \quad P = 2$
Q090 $S = \frac{7}{3}, \quad P = -2$
Q091 C
Q092 A
Q093 B
Q094 A
Q095 12

Q096 $\frac{81}{2}$

Q097 1

Q098 -1

Q099 -2

Q100 -10

Q101 $\frac{14}{3}$

Q102 -10

Q103 A

Q104 B

Q105 A

Q106 B

Q107 D

Q108 B

Q109 B

Q110 D

Q111 -4

Q112 $\frac{2}{3}$

Q113 $S = 0, \quad P = -\frac{3}{5}$

Q114 $\frac{2}{3}$

Q115 6

Q116 C

Q117 D

Q118 D

Q119 D

Q120 D

Q121 $\frac{5}{2}$

Q122 $\frac{1}{2}$

Q123 11

Q124 $\frac{23}{3}$

Q125 A

Q126 C

Q127 D

Q128 D

Q129 $k = 6, \quad x = 2, 6$

Q130 $m = 8, \quad x = -2, -4$

Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Q132 A

Q133 A

Q134 D

Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Q137 $m = 5, \quad x = -5, -1$

Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Q139 D

Q140 A

Q141 B

Q142 B

Q143 $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

Q144 A

Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Q146 B

Q147 $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$

Q148 $(x + 1 - 2i)(x + 1 + 2i)$

Q149 $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$

Q150 $(x + 2 - i)(x + 2 + i)$

Q151 $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

Q152 $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

Q153 $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

Q154 $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$

Q156 C

Q157 A

Q158 D

Q159 D

Q160 B

Q161 A

Q162 C

Q163 A

Q164 B

Q165 $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$

Q166 $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$

Q167 $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$

Q168 $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$

Q169 B

Q170 B

Q171 D

Q172 C

Q173 $2x^2 - 2x + 1 = 0$

Q174 $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

Q175 $x^2 - 6x + 13 = 0$

Q176 $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$

Q177 $x^2 - x - 2 = 0$

Q178 $x^2 - 2x + 17 = 0$

Q179 $3x^2 - 10x + 6 = 0$

Q180 $1 + 2i, \ 1 - 2i$

Q181 $2 + \sqrt{2}, \ 2 - \sqrt{2}$

Q182 B

Q183 B

Q184 B

Q185 A

Q186 A

Q187 D

Q188 A

Q189 B

Q190 B

Q191 $x^2 - 2x - 4 = 0$

Q192 C

Q193 $x^2 - 7x + 9 = 0$

Q194 $x^2 + 10x + 49 = 0$

Q195 D

Q196 C

Q197 $4x^2 + 3x + 2 = 0$

Q198 A

Q199 $x^2 + 8x + 24 = 0$

Q200 D

Q201 $-2 < m < -1$

Q202 $-1 < m < -\frac{1}{2}$

Q203 A

Q204 B

Q205 $-6 < m < -2$

Q206 $k > 10$

Q207 C

Q208 B

Q209 $m > -4$

Q210 $k > -3$

Q211 A

Q212 D

Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c; other roots a and b.

Q221 $x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$

Q222 For $P = \alpha\beta \neq 0$: $\quad Px^2 - Sx + 1 = 0$

Q223 $-4 < m < -3$

Q224 $m > 5$

Q225 $m < -4$

Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$

Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$P(a) = R$ when divisor $= x - a$

Factor

$P(a) = 0 \Rightarrow (x - a) \mid P(x)$

Roots

$P(x) = a \prod (x - r_i)$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Polynomial Theorems**

Q001 Written

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.

Solution

1. $Use x = -2$
2. $P(-2) = 2$
3. Therefore the final answer is 2.

Final answer

2

Q002 Written

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.

Solution

1. $Use x = 1$
2. $P(1) = 6$
3. Therefore the final answer is 6.

Final answer

6

Worked Solutions **Polynomial Theorems**

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 2.

Correct option: C

Final answer

2

Q004 MCQ

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 6.

Correct option: C

Final answer

6

Worked Solutions **Polynomial Theorems**

Q005 Written

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.

Solution

1. $Usex = 2/3$
2. $EvaluateP(2/3) = 8$
3. Therefore the final answer is 8.

Final answer

8

Q006 Written

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.

Solution

1. $Usex = 1/2$
2. $P(1/2) = -1/8$
3. Therefore the final answer is $-\frac{1}{8}$.

Final answer

$-\frac{1}{8}$

Worked Solutions **Polynomial Theorems**

Q007 Written

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.

Solution

1. $Use x = 2/3$
2. $P(2/3) = 0$
3. Therefore the final answer is 0.

Final answer

0

Q008 Written

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.

Solution

1. $Use x = -1/2$
2. $P(-1/2) = 4$
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions **Polynomial Theorems**

Q009 *MCQ*

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 8.

Correct option: D

Final answer

8

Q010 *MCQ*

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-\frac{1}{8}$.

Correct option: A

Final answer

$-\frac{1}{8}$

Worked Solutions **Polynomial Theorems**

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 0.

Correct option: D

Final answer

0

Q012 MCQ

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 4.

Correct option: A

Final answer

4

Q013 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a.

Solution

1. $P(-3) = 4$
2. $9a - 35 = 4, hencea = 13/3$
3. Therefore the final answer is $a = 13/3$.

Final answer

$a = 13/3$

Q014 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a.

Solution

1. $P(3) = 1$
2. $9a + 37 = 1, hencea = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$a = -4$

Worked Solutions **Polynomial Theorems**

Q015 Written

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2, find a.

Solution

1. $P(-1) = -2$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$a = -4$

Q016 Written

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a.

Solution

1. $P(-3) = 8$
2. $a = -16$
3. Therefore the final answer is $a = -16$.

Final answer

$a = -16$

Worked Solutions **Polynomial Theorems**

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = 13/3$.

Correct option: B

Final answer

$a = 13/3$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: B

Final answer

$a = -4$

Worked Solutions **Polynomial Theorems**

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: C

Final answer

$a = -4$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -16$.

Correct option: A

Final answer

$a = -16$

Worked Solutions **Polynomial Theorems**

Q021 Written

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a.

Solution

1. $P(2) = 0$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$a = -4$

Q022 MCQ

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: A

Final answer

$a = -4$

Worked Solutions **Polynomial Theorems**

Q023 **Written**

A polynomial $P(x)$ leaves remainders 3 on division by $x-1$ and -5 on division by $x+3$. Find the remainder on division by x^2+2x-3.

Solution

1. $Let R = ax + b$
2. $a + b = 3 and - 3a + b = -5$
3. $R = 2x + 1$
4. Therefore the final answer is $2x + 1$.

Final answer

$2x + 1$

Q024 **Written**

A polynomial $P(x)$ leaves remainders 3 on division by $x-2$ and -7 on division by $x+3$. Find the remainder on division by x^2+x-6.

Solution

1. $Let R = ax + b$
2. $2a + b = 3 and - 3a + b = -7$
3. $R = -10x + 13$
4. Therefore the final answer is $-10x + 13$.

Final answer

$-10x + 13$

Worked Solutions **Polynomial Theorems**

Q025 Written

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$.

Solution

1. $Let R = ax + b$
2. $a + b = 5and - 2a + b = -1$
3. $R = 2x + 3$
4. Therefore the final answer is $2x + 3$.

Final answer

$2x + 3$

Q026 Written

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. $Let R = ax + b$
2. $2a + b = 4and - 3a + b = -11$
3. $R = -15x + 19$
4. Therefore the final answer is $-15x + 19$.

Final answer

$-15x + 19$

Worked Solutions **Polynomial Theorems**

Q027 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x-1$ and -5 on division by $x+3$. Find the remainder on division by x^2+2x-3 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $2x+1$.

Correct option: A

Final answer

$2x+1$

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x-2$ and -7 on division by $x+3$. Find the remainder on division by x^2+x-6 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-10x+13$.

Correct option: A

Final answer

$-10x+13$

Worked Solutions **Polynomial Theorems**

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x-1$ and -1 on division by $x+2$. Find the remainder on division by $(x-1)(x+2)$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $2x+3$.

Correct option: A

Final answer

$2x+3$

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x-2$ and -11 on division by $x+3$. Find the remainder on division by x^2+x-6 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-15x+19$.

Correct option: A

Final answer

$-15x+19$

Worked Solutions **Polynomial Theorems**

Q031 Written

A polynomial $P(x)$ leaves remainder $7x+6$ on division by x^2-x-6, and $4x-3$ on division by x^2+2x-3. Find the remainder on division by x^2+x-2.

Solution

1. $LetR = ax + b$
2. $Usesharedroots - 2and1 : -2a + b = -8anda + b = 1$
3. $R = 3x - 2$
4. Therefore the final answer is $3x - 2$.

Final answer

$3x - 2$

Q032 Written

A polynomial $P(x)$ leaves remainder $3x+2$ on division by x^2-x-2, and $7x-6$ on division by x^2+2x-3. Find the remainder on division by x^2-1.

Solution

1. $LetR = ax + b$
2. $At - 1and1 : -a + b = -1anda + b = 1$
3. $R = x$
4. Therefore the final answer is x.

Final answer

x

Worked Solutions **Polynomial Theorems**

Q033 Written

A polynomial $P(x)$ leaves remainder $2x-2$ on division by x^2-2x-3, and $-x+7$ on division by x^2-4. Find the remainder on division by x^2-x-2.

Solution

1. Use shared roots -1 and 2
2. $-a+b=-4 and 2a+b=5, so R=3x-1$
3. Therefore the final answer is $3x-1$.

Final answer

$3x-1$

Q034 Written

A polynomial $P(x)$ leaves remainder $-x+4$ on division by x^2-3x+2, and $3x$ on division by x^2-4x+3. Find the remainder on division by x^2-5x+6.

Solution

1. Use shared roots 2 and 3
2. $2a+b=2 and 3a+b=9, so R=7x-12$
3. Therefore the final answer is $7x-12$.

Final answer

$7x-12$

Worked Solutions **Polynomial Theorems**

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x+6$ on division by x^2-x-6, and $4x-3$ on division by x^2+2x-3. Find the remainder on division by x^2+x-2 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $3x-2$.

Correct option: B

Final answer

$3x-2$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x+2$ on division by x^2-x-2, and $7x-6$ on division by x^2+2x-3. Find the remainder on division by x^2-1 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is x.

Correct option: B

Final answer

x

Worked Solutions **Polynomial Theorems**

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $3x - 1$.

Correct option: A

Final answer

$3x - 1$

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $7x - 12$.

Correct option: D

Final answer

$7x - 12$

Worked Solutions **Polynomial Theorems**

Q039 Written

A polynomial $P(x)$ leaves remainder $x-5$ on division by $(x+1)^2$, and 6 on division by $x-2$. Find the remainder on division by $(x+1)^2(x-2)$.

Solution

1. $LetR = ax^2 + bx + c$
2. $R(-1) = -6, R'(-1) = 1, R(2) = 6$
3. $SolvetogetR = x^2 + 3x - 4$
4. Therefore the final answer is $x^2 + 3x - 4$.

Final answer

$x^2 + 3x - 4$

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x-5$ on division by $(x+1)^2$, and 6 on division by $x-2$. Find the remainder on division by $(x+1)^2(x-2)$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x^2 + 3x - 4$.

Correct option: A

Final answer

$x^2 + 3x - 4$

Worked Solutions **Polynomial Theorems**

Q041 Written

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Solution

1. Use -2 and coefficients 1,0,5,11
2. Bottom row 1,-2,9,-7
3. Therefore the final answer is $Q = x^2 - 2x + 9, R = -7$.

Final answer

$Q = x^2 - 2x + 9, R = -7$

Q042 MCQ

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 - 2x + 9, R = -7$.

Correct option: A

Final answer

$Q = x^2 - 2x + 9, R = -7$

Worked Solutions **Polynomial Theorems**

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,0,5,-6
2. Bottom row 1,1,6,0
3. Therefore the final answer is $Q = x^2 + x + 6, R = 0$.

Final answer

$Q = x^2 + x + 6, R = 0$

Q044 Written

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Solution

1. Use 3 and coefficients 3,-5,-3,-1
2. Bottom row 3,4,9,26
3. Therefore the final answer is $Q = 3x^2 + 4x + 9, R = 26$.

Final answer

$Q = 3x^2 + 4x + 9, R = 26$

Worked Solutions **Polynomial Theorems**

Q045 **Written**

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Solution

1. Use 2 and coefficients 1,0,-5,6
2. Bottom row 1,2,-1,4
3. Therefore the final answer is $Q = x^2 + 2x - 1, R = 4$.

Final answer

$Q = x^2 + 2x - 1, R = 4$

Q046 **Written**

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,2,-1,-3
2. Bottom row 1,3,2,-1
3. Therefore the final answer is $Q = x^2 + 3x + 2, R = -1$.

Final answer

$Q = x^2 + 3x + 2, R = -1$

Worked Solutions **Polynomial Theorems**

Q047 Written

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 4,-2,2
2. Bottom row 4,2,4
3. Therefore the final answer is $Q = 4x + 2, R = 4$.

Final answer

$Q = 4x + 2, R = 4$

Q048 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + x + 6, R = 0$.

Correct option: B

Final answer

$Q = x^2 + x + 6, R = 0$

Worked Solutions **Polynomial Theorems**

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 3x^2 + 4x + 9, R = 26$.

Correct option: D

Final answer

$Q = 3x^2 + 4x + 9, R = 26$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + 2x - 1, R = 4$.

Correct option: D

Final answer

$Q = x^2 + 2x - 1, R = 4$

Worked Solutions **Polynomial Theorems**

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + 3x + 2, R = -1$.

Correct option: A

Final answer

$Q = x^2 + 3x + 2, R = -1$

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 4x + 2, R = 4$.

Correct option: B

Final answer

$Q = 4x + 2, R = 4$

Worked Solutions **Polynomial Theorems**

Q053 Written

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Solution

1. Use -2 and coefficients 3,5,0,-2,-12
2. Bottom row 3,-1,2,-6,0
3. Therefore the final answer is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Final answer

$Q = 3x^3 - x^2 + 2x - 6, R = 0$

Q054 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Correct option: D

Final answer

$Q = 3x^3 - x^2 + 2x - 6, R = 0$

Worked Solutions **Polynomial Theorems**

Q055 **Written**

Factorise $x^3 + 2x^2 - x - 2$.

Solution

1. $P(1) = 0; divide by x - 1$
2. $Quotient x^2 + 3x + 2 = (x + 1)(x + 2)$
3. Therefore the final answer is $(x - 1)(x + 1)(x + 2)$.

Final answer

$(x - 1)(x + 1)(x + 2)$

Q056 **Written**

Factorise $x^3 - 2x^2 - 5x + 6$.

Solution

1. $P(1) = 0; quotient x^2 - x - 6 = (x - 3)(x + 2)$
2. Completed factorisation
3. Therefore the final answer is $(x - 1)(x - 3)(x + 2)$.

Final answer

$(x - 1)(x - 3)(x + 2)$

Worked Solutions **Polynomial Theorems**

Q057 Written

Factorise $x^3 - x^2 - 16x - 20$.

Solution

1. $P(-2) = 0; dividebyx + 2twice$
2. Remaining x-5
3. Therefore the final answer is $(x + 2)^2(x - 5)$.

Final answer

$(x + 2)^2(x - 5)$

Q058 Written

Factorise $2x^3 - x^2 - 5x - 2$.

Solution

1. $P(2) = 0; quotient 2x^2 + 3x + 1$
2. Therefore the final answer is $(x - 2)(2x + 1)(x + 1)$.

Final answer

$(x - 2)(2x + 1)(x + 1)$

Worked Solutions **Polynomial Theorems**

Q059 **Written**

Factorise $x^3 - 7x - 6$.

Solution

1. Test divisor 3 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x-3)(x+1)(x+2)$.

Final answer

$(x-3)(x+1)(x+2)$

Q060 **Written**

Factorise $x^3 + 4x^2 + x - 6$.

Solution

1. Test divisor 1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x-1)(x+2)(x+3)$.

Final answer

$(x-1)(x+2)(x+3)$

Worked Solutions **Polynomial Theorems**

Q061 **Written**

Factorise $x^3 + 6x^2 + 11x + 6$.

Solution

1. Test divisor -1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x + 1)(x + 2)(x + 3)$.

Final answer

$(x + 1)(x + 2)(x + 3)$

Q062 **Written**

Factorise $x^3 - 3x + 2$.

Solution

1. Test divisor 1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 1)^2(x + 2)$.

Final answer

$(x - 1)^2(x + 2)$

Worked Solutions **Polynomial Theorems**

Q063 **Written**

Factorise $x^3 + x^2 - 8x - 12$.

Solution

1. Test divisor -2 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 3)(x + 2)^2$.

Final answer

$(x - 3)(x + 2)^2$

Q064 **Written**

Factorise $x^3 - 3x^2 - 9x - 5$.

Solution

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x + 1)^2$.

Final answer

$(x - 5)(x + 1)^2$

Worked Solutions **Polynomial Theorems**

Q065 **Written**

Factorise $2x^3 - 3x^2 - 3x + 2$**.**

Solution

1. Test divisor 2 and divide
2. $Factorquotient 2x^2 + x - 1$
3. Therefore the final answer is $(x - 2)(x + 1)(2x - 1)$.

Final answer

$(x - 2)(x + 1)(2x - 1)$

Q066 **Written**

Factorise $3x^3 + x^2 - 8x + 4$**.**

Solution

1. Test divisor 1 and divide
2. $Factorquotient 3x^2 + 4x - 4$
3. Therefore the final answer is $(x - 1)(x + 2)(3x - 2)$.

Final answer

$(x - 1)(x + 2)(3x - 2)$

Worked Solutions **Polynomial Theorems**

Q067 Written

Factorise $2x^3 + 7x^2 + 8x + 3$.

Solution

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x + 1)^2(2x + 3)$.

Final answer

$(x + 1)^2(2x + 3)$

Q068 MCQ

Factorise $x^3 + 2x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x + 1)(x + 2)$.

Correct option: D

Final answer

$(x - 1)(x + 1)(x + 2)$

Worked Solutions **Polynomial Theorems**

Q069 *MCQ*

Factorise $x^3 - 2x^2 - 5x + 6$ **Select the correct answer:**

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x-3)(x+2)$.

Correct option: D

Final answer

$(x-1)(x-3)(x+2)$

Q070 *MCQ*

Factorise $x^3 - x^2 - 16x - 20$ **Select the correct answer:**

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x+2)^2(x-5)$.

Correct option: C

Final answer

$(x+2)^2(x-5)$

Q071 MCQ

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)(2x+1)(x+1)$.

Correct option: B

Final answer

$(x-2)(2x+1)(x+1)$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-3)(x+1)(x+2)$.

Correct option: B

Final answer

$(x-3)(x+1)(x+2)$

Worked Solutions **Polynomial Theorems**

Q073 *MCQ*

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x + 2)(x + 3)$.

Correct option: C

Final answer

$(x - 1)(x + 2)(x + 3)$

Q074 *MCQ*

Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x + 1)(x + 2)(x + 3)$.

Correct option: B

Final answer

$(x + 1)(x + 2)(x + 3)$

Worked Solutions **Polynomial Theorems**

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)^2(x+2)$.

Correct option: D

Final answer

$(x-1)^2(x+2)$

Q076 MCQ

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-3)(x+2)^2$.

Correct option: D

Final answer

$(x-3)(x+2)^2$

Worked Solutions **Polynomial Theorems**

Q077 MCQ

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-5)(x+1)^2$.

Correct option: D

Final answer

$(x-5)(x+1)^2$

Q078 MCQ

Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)(x+1)(2x-1)$.

Correct option: A

Final answer

$(x-2)(x+1)(2x-1)$

Worked Solutions **Polynomial Theorems**

Q079 MCQ

Factorise $3x^3 + x^2 - 8x + 4$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x+2)(3x-2)$.

Correct option: C

Final answer

$(x-1)(x+2)(3x-2)$

Q080 MCQ

Factorise $2x^3 + 7x^2 + 8x + 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x+1)^2(2x+3)$.

Correct option: D

Final answer

$(x+1)^2(2x+3)$

Worked Solutions **Polynomial Theorems**

Q081 Written

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$.

Solution

1. $P(2) = 0; dividebyx - 2twice$
2. $Remainingquadraticx^2 - 2x - 4$
3. Therefore the final answer is $(x-2)^2(x^2 - 2x - 4)$.

Final answer

$(x-2)^2(x^2 - 2x - 4)$

Q082 Written

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$.

Solution

1. Test roots and divide successively
2. Four linear factors
3. Therefore the final answer is $(x-3)(x-2)(x+1)(x+7)$.

Final answer

$(x-3)(x-2)(x+1)(x+7)$

Worked Solutions **Polynomial Theorems**

Q083 **Written**

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$.

Solution

1. $P(1) = 0; dividetwice$
2. $Remainingquadraticx^2 - 2x + 3$
3. Therefore the final answer is $(x-1)^2(x^2-2x+3)$.

Final answer

$(x-1)^2(x^2-2x+3)$

Q084 **Written**

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$.

Solution

1. Test roots 1,2,3,4 successively
2. Four linear factors
3. Therefore the final answer is $(x-4)(x-3)(x-2)(x-1)$.

Final answer

$(x-4)(x-3)(x-2)(x-1)$

Worked Solutions **Polynomial Theorems**

Q085 **Written**

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$.

Solution

1. Test roots 1 and -2 successively
2. Completed factorisation
3. Therefore the final answer is $(x-5)(x-1)^2(x+2)$.

Final answer

$(x-5)(x-1)^2(x+2)$

Q086 **Written**

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$.

Solution

1. Test roots 2,-2,4,-3/2
2. Completed factorisation
3. Therefore the final answer is $(x-4)(x-2)(x+2)(2x+3)$.

Final answer

$(x-4)(x-2)(x+2)(2x+3)$

Worked Solutions **Polynomial Theorems**

Q087 Written

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$.

Solution

1. Test roots 1,-2
2. Quadratic quotient remains
3. Therefore the final answer is $(x-1)(x+2)(x^2-5x+3)$.

Final answer

$(x-1)(x+2)(x^2-5x+3)$

Q088 Written

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$.

Solution

1. Test roots 3,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x-3)(x+1)(x^2-2x+2)$.

Final answer

$(x-3)(x+1)(x^2-2x+2)$

Worked Solutions **Polynomial Theorems**

Q089 Written

Factorise $x^4 - x^3 + x^2 + x - 2$.

Solution

1. Test roots 1,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x-1)(x+1)(x^2-x+2)$.

Final answer

$(x-1)(x+1)(x^2-x+2)$

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)^2(x^2-2x-4)$.

Correct option: D

Final answer

$(x-2)^2(x^2-2x-4)$

Worked Solutions **Polynomial Theorems**

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-3)(x-2)(x+1)(x+7)$.

Correct option: A

Final answer

$(x-3)(x-2)(x+1)(x+7)$

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)^2(x^2-2x+3)$.

Correct option: D

Final answer

$(x-1)^2(x^2-2x+3)$

Worked Solutions **Polynomial Theorems**

Q093 MCQ

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ **Select the correct answer:**

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-4)(x-3)(x-2)(x-1)$.

Correct option: D

Final answer

$(x-4)(x-3)(x-2)(x-1)$

Q094 MCQ

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ **Select the correct answer:**

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-5)(x-1)^2(x+2)$.

Correct option: C

Final answer

$(x-5)(x-1)^2(x+2)$

Worked Solutions **Polynomial Theorems**

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-4)(x-2)(x+2)(2x+3)$.

Correct option: D

Final answer

$(x-4)(x-2)(x+2)(2x+3)$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x+2)(x^2-5x+3)$.

Correct option: A

Final answer

$(x-1)(x+2)(x^2-5x+3)$

Worked Solutions **Polynomial Theorems**

Q097 MCQ

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-3)(x+1)(x^2-2x+2)$.

Correct option: B

Final answer

$(x-3)(x+1)(x^2-2x+2)$

Q098 MCQ

Factorise $x^4 - x^3 + x^2 + x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x+1)(x^2-x+2)$.

Correct option: C

Final answer

$(x-1)(x+1)(x^2-x+2)$

Worked Solutions

Polynomial Theorems

Q099 Written

Factorise $2x^3 - 3x^2 - x - 2$.

Solution

1. $P(2) = 0; quotient 2x^2 + x + 1 has discriminant - 7$
2. Therefore the final answer is $(x - 2)(2x^2 + x + 1)$.

Final answer

$(x - 2)(2x^2 + x + 1)$

Q100 Written

Factorise $2x^3 - 7x^2 + 3x + 6$.

Solution

1. Test divisor 2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 2)(2x^2 - 3x - 3)$.

Final answer

$(x - 2)(2x^2 - 3x - 3)$

Worked Solutions **Polynomial Theorems**

Q101 Written

Factorise $x^3 + 4x^2 + 2x - 4$.

Solution

1. Test divisor -2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x + 2)(x^2 + 2x - 2)$.

Final answer

$(x + 2)(x^2 + 2x - 2)$

Q102 Written

Factorise $x^3 - 3x^2 + 2$.

Solution

1. Test divisor 1 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x^2 - 2x - 2)$.

Final answer

$(x - 1)(x^2 - 2x - 2)$

Worked Solutions **Polynomial Theorems**

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)(2x^2+x+1)$.

Correct option: C

Final answer

$(x-2)(2x^2+x+1)$

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)(2x^2-3x-3)$.

Correct option: B

Final answer

$(x-2)(2x^2-3x-3)$

Worked Solutions **Polynomial Theorems**

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x + 2)(x^2 + 2x - 2)$.

Correct option: B

Final answer

$(x + 2)(x^2 + 2x - 2)$

Q106 MCQ

Factorise $x^3 - 3x^2 + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x^2 - 2x - 2)$.

Correct option: D

Final answer

$(x - 1)(x^2 - 2x - 2)$

Worked Solutions **Polynomial Theorems**

Q107 **Challenge**

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.

Solution

1. $Substitute x = -1$
2. $(-1)^{57} + 57 = 56$
3. Therefore the final answer is 56.

Final answer

56

Worked Solutions **Polynomial Theorems**

Q108 **Challenge**

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a, then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Solution

1. Use the divisor zeros: $P(-1/2) = 0$ and $P(-2) = 0$.
2. Hence $a = 10$ for $2x + 1$, and $a = -11$ for $x + 2$.
3. Each linear factor corresponds to a polynomial root.

Final answer

$a = 10 \quad \text{for } 2x + 1$

$a = -11 \quad \text{for } x + 2$

linear factors $\leftrightarrow$ roots

Q109 Challenge

A polynomial $P(x)$ is divisible by $x-1$ and leaves remainder -4 on division by $x+3$. Find the remainder on division by x^2+2x-3.

Solution

1. $a+b=0and-3a+b=-4$
2. $R=x-1$
3. Therefore the final answer is $x-1$.

Final answer

$x-1$

Worked Solutions **Polynomial Theorems**

Q110 **Challenge**

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Solution

1. $P = (2x - 1)^2(x^2 - 3)$
2. $Remainingrootsare + / - \sqrt(3)$
3. Therefore the final answer is $x = \sqrt{3}, -\sqrt{3}$.

Final answer

$x = \sqrt{3}, -\sqrt{3}$

Q111 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b.

Solution

1. $P(1) = 0 gives a + b = -10$
2. $P(-2) = 0 gives 2a - b = -26$
3. $a = -12, b = 2$
4. Therefore the final answer is $a = -12, b = 2$.

Final answer

$a = -12, b = 2$

Q112 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$.

Solution

1. $P = (x-1)(x+2)(x^2-4x-6)$
2. $Solvequadratic : x = 2 \pm \sqrt{(10)}$
3. $AllfourrootssatisfyP = 0$
4. Therefore the final answer is $x = 1, -2, 2 \pm \sqrt{10}$.

Final answer

$x = 1, -2, 2 \pm \sqrt{10}$

Q113 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 56.

Correct option: B

Final answer

56

Worked Solutions **Polynomial Theorems**

Q114 **Challenge**

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a, then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.

$$a = 10 \quad \text{for } 2x + 1$$

2. The correct value is $a = -11 \quad \text{for } x + 2$.

$$\text{linear factors} \leftrightarrow \text{roots}$$

Correct option: D

Final answer

$a = 10 \quad \text{for } 2x + 1$

$a = -11 \quad \text{for } x + 2$

linear factors $\leftrightarrow$ roots

Worked Solutions **Polynomial Theorems**

Q115 **Challenge**

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x - 1$.

Correct option: C

Final answer

$x - 1$

Worked Solutions **Polynomial Theorems**

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x = \sqrt{3}, -\sqrt{3}$.

Correct option: A

Final answer

$x = \sqrt{3}, -\sqrt{3}$

Q117 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -12, b = 2$.

Correct option: C

Final answer

$a = -12, b = 2$

Worked Solutions **Polynomial Theorems**

Q118 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x = 1, -2, 2 \pm \sqrt{10}$.

Correct option: A

Final answer

$x = 1, -2, 2 \pm \sqrt{10}$

Worked Solutions **Polynomial Theorems**

Quiz MCQ 1 **Concept Quiz · MCQ**

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 9.

Correct option: C

Final answer

9

Quiz MCQ 2 **Concept Quiz · MCQ**

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $P(-1) = 0$.

Correct option: C

Final answer

$P(-1) = 0$

Worked Solutions **Polynomial Theorems**

Quiz MCQ 3 Concept Quiz · MCQ

The remainder on division by a quadratic divisor has the form Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $ax + b$.

Correct option: B

Final answer

$ax + b$

Quiz MCQ 4 Concept Quiz · MCQ

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Syntheticdivision$.

Correct option: C

Final answer

$Syntheticdivision$

Worked Solutions **Polynomial Theorems**

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: 4

Final answer

4

Quiz WRITTEN 6 Concept Quiz · Written

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $R = 3x$

Final answer

$R = 3x$

Worked Solutions **Polynomial Theorems**

Quiz WRITTEN 7 Concept Quiz · Written

Factorise $x^3 - 6x^2 + 11x - 6$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $(x-1)(x-2)(x-3)$

Final answer

$(x-1)(x-2)(x-3)$

Quiz WRITTEN 8 Concept Quiz · Written

If $x-1$ and $x+2$ are factors of $x^4 - 3x^3 + ax^2 + bx + 12$, find a, b.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $a = -12, b = 2$

Final answer

$a = -12, b = 2$

Answer key

Use the question number shown on each page.

Q001 2
Q002 6
Q003 C
Q004 C
Q005 8
Q006 $-\frac{1}{8}$
Q007 0
Q008 4
Q009 D
Q010 A
Q011 D
Q012 A
Q013 $a = 13/3$
Q014 $a = -4$
Q015 $a = -4$
Q016 $a = -16$
Q017 B
Q018 B
Q019 C
Q020 A
Q021 $a = -4$
Q022 A
Q023 $2x + 1$
Q024 $-10x + 13$
Q025 $2x + 3$
Q026 $-15x + 19$
Q027 A
Q028 A
Q029 A
Q030 A
Q031 $3x - 2$
Q032 x
Q033 $3x - 1$
Q034 $7x - 12$
Q035 B
Q036 B
Q037 A
Q038 D
Q039 $x^2 + 3x - 4$
Q040 A
Q041 $Q = x^2 - 2x + 9, R = -7$
Q042 A
Q043 $Q = x^2 + x + 6, R = 0$
Q044 $Q = 3x^2 + 4x + 9, R = 26$
Q045 $Q = x^2 + 2x - 1, R = 4$
Q046 $Q = x^2 + 3x + 2, R = -1$
Q047 $Q = 4x + 2, R = 4$
Q048 B
Q049 D
Q050 D
Q051 A
Q052 B
Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$
Q054 D
Q055 $(x-1)(x+1)(x+2)$
Q056 $(x-1)(x-3)(x+2)$
Q057 $(x+2)^2(x-5)$
Q058 $(x-2)(2x+1)(x+1)$
Q059 $(x-3)(x+1)(x+2)$
Q060 $(x-1)(x+2)(x+3)$
Q061 $(x+1)(x+2)(x+3)$
Q062 $(x-1)^2(x+2)$
Q063 $(x-3)(x+2)^2$
Q064 $(x-5)(x+1)^2$
Q065 $(x-2)(x+1)(2x-1)$
Q066 $(x-1)(x+2)(3x-2)$
Q067 $(x+1)^2(2x+3)$
Q068 D
Q069 D
Q070 C
Q071 B
Q072 B
Q073 C
Q074 B
Q075 D
Q076 D
Q077 D
Q078 A
Q079 C
Q080 D
Q081 $(x-2)^2(x^2-2x-4)$
Q082 $(x-3)(x-2)(x+1)(x+7)$
Q083 $(x-1)^2(x^2-2x+3)$
Q084 $(x-4)(x-3)(x-2)(x-1)$
Q085 $(x-5)(x-1)^2(x+2)$
Q086 $(x-4)(x-2)(x+2)(2x+3)$
Q087 $(x-1)(x+2)(x^2-5x+3)$
Q088 $(x-3)(x+1)(x^2-2x+2)$
Q089 $(x-1)(x+1)(x^2-x+2)$
Q090 D
Q091 A
Q092 D
Q093 D
Q094 C
Q095 D
Q096 A
Q097 B
Q098 C
Q099 $(x-2)(2x^2+x+1)$
Q100 $(x-2)(2x^2-3x-3)$
Q101 $(x+2)(x^2+2x-2)$
Q102 $(x-1)(x^2-2x-2)$
Q103 C
Q104 B

Q105 B

Q106 D

Q107 56

$a = 10$ for $2x + 1$

Q108 $a = -11$ for $x + 2$

linear factors ↔ roots

Q109 $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B

Q114 D

Q115 C

Q116 A

Q117 C

Q118 A

Quiz MCQ 1 C

Quiz MCQ 2 C

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 4

Quiz WRITTEN 6 $R = 3x$

Quiz WRITTEN 7 $(x-1)(x-2)(x-3)$

Quiz WRITTEN 8 $a = -12, b = 2$

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$P(a) = 0 \Rightarrow x - a$ is a factor

Degree

$\deg(PQ) = \deg P + \deg Q$

Conjugate roots

$a + bi \Rightarrow a - bi$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Higher-Degree Equations**

Q001 **Written**

$x^3 - 8 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$.

Final answer

$x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

Q002 **Written**

$x^3 = -1$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$.

Final answer

$x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q003 **Written**

$x^3 = 27$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$.

Final answer

$x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$

Q004 **Written**

$x^3 = -125$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$.

Final answer

$x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q005 **Written**

$8x^3 = 1$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$.

Final answer

$x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$

Q006 **MCQ**

$x^3 - 8 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$.

Correct option: B

Final answer

$x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

Worked Solutions **Higher-Degree Equations**

Q007 MCQ

$x^3 = -1$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$.

Correct option: D

Final answer

$x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$

Q008 MCQ

$x^3 = 27$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$.

Correct option: B

Final answer

$x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q009 *MCQ*

$x^3 = -125$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$.

Correct option: A

Final answer

$x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$

Q010 *MCQ*

$8x^3 = 1$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$.

Correct option: C

Final answer

$x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$

Worked Solutions **Higher-Degree Equations**

Q011 **Written**

$x^3 - 2x^2 - 5x + 6 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 1, 3$.

Final answer

$x = -2, 1, 3$

Q012 **Written**

$3x^3 + x^2 - 8x + 4 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 2/3, 1$.

Final answer

$x = -2, 2/3, 1$

Worked Solutions **Higher-Degree Equations**

Q013 **Written**

$2x^3 - 3x^2 - x - 2 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$.

Final answer

$x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$

Q014 **Written**

$x^3 - x^2 - 16x - 20 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{repeated}), 5$.

Final answer

$x = -2(\text{repeated}), 5$

Worked Solutions **Higher-Degree Equations**

Q015 **Written**

$x^3 + 6x^2 + 11x + 6 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, -1$.

Final answer

$x = -3, -2, -1$

Q016 **Written**

$x^3 + 3x^2 - 4x - 12 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, 2$.

Final answer

$x = -3, -2, 2$

Worked Solutions **Higher-Degree Equations**

Q017 Written

$3x^3 - 2x^2 - 3x + 2 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 2/3, 1$.

Final answer

$x = -1, 2/3, 1$

Q018 Written

$x^3 - x^2 - 4 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/2$.

Final answer

$x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q019 Written

$x^3 + 4x^2 + 2x - 4 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Final answer

$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

Q020 Written

$x^3 - 4x^2 + 9x - 10 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, 1 - 2\,i, 1 + 2\,i$.

Final answer

$x = 2, 1 - 2\,i, 1 + 2\,i$

Worked Solutions **Higher-Degree Equations**

Q021 Written

$x^3 + x^2 - 8x - 12 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{repeated}), 3$.

Final answer

$x = -2(\text{repeated}), 3$

Q022 Written

$2x^3 + 7x^2 + 8x + 3 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1(\text{repeated}), -3/2$.

Final answer

$x = -1(\text{repeated}), -3/2$

Worked Solutions **Higher-Degree Equations**

Q023 Written

$x^3 + 6x^2 + 12x + 8 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{triple})$.

Final answer

$x = -2(\text{triple})$

Q024 MCQ

$x^3 - 2x^2 - 5x + 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, 1, 3$.

Correct option: C

Final answer

$x = -2, 1, 3$

Worked Solutions **Higher-Degree Equations**

Q025 MCQ

$3x^3 + x^2 - 8x + 4 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, 2/3, 1$.

Correct option: D

Final answer

$x = -2, 2/3, 1$

Q026 MCQ

$2x^3 - 3x^2 - x - 2 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$.

Correct option: D

Final answer

$x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$

Worked Solutions **Higher-Degree Equations**

Q027 MCQ

$x^3 - x^2 - 16x - 20 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2(\text{repeated}), 5$.

Correct option: B

Final answer

$x = -2(\text{repeated}), 5$

Q028 MCQ

$x^3 + 6x^2 + 11x + 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, -2, -1$.

Correct option: C

Final answer

$x = -3, -2, -1$

Worked Solutions **Higher-Degree Equations**

Q029 MCQ

$x^3 + 3x^2 - 4x - 12 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, -2, 2$.

Correct option: D

Final answer

$x = -3, -2, 2$

Q030 MCQ

$3x^3 - 2x^2 - 3x + 2 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 2/3, 1$.

Correct option: D

Final answer

$x = -1, 2/3, 1$

Worked Solutions **Higher-Degree Equations**

Q031 MCQ

$x^3 - x^2 - 4 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}\, i)/2, (-1 + \sqrt{7}\, i)/2$.

Correct option: A

Final answer

$x = 2, (-1 - \sqrt{7}\, i)/2, (-1 + \sqrt{7}\, i)/2$

Worked Solutions **Higher-Degree Equations**

Q032 MCQ

$x^3 + 4x^2 + 2x - 4 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Correct option: D

Final answer

$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

Worked Solutions **Higher-Degree Equations**

Q033 MCQ

$x^3 - 4x^2 + 9x - 10 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, 1 - 2i, 1 + 2i$.

Correct option: D

Final answer

$x = 2, 1 - 2i, 1 + 2i$

Q034 MCQ

$x^3 + x^2 - 8x - 12 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2(\text{repeated}), 3$.

Correct option: D

Final answer

$x = -2(\text{repeated}), 3$

Worked Solutions **Higher-Degree Equations**

Q035 MCQ

$2x^3 + 7x^2 + 8x + 3 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1(\text{repeated}), -3/2$.

Correct option: B

Final answer

$x = -1(\text{repeated}), -3/2$

Q036 MCQ

$x^3 + 6x^2 + 12x + 8 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2(\text{triple})$.

Correct option: D

Final answer

$x = -2(\text{triple})$

Worked Solutions   **Higher-Degree Equations**

Q037 **Written**

$x^4 - 7x^2 - 18 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$.

Final answer

$x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$

Q038 **Written**

$x^4 - 5x^2 + 4 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1, 1, 2$.

Final answer

$x = -2, -1, 1, 2$

Worked Solutions **Higher-Degree Equations**

Q039 **Written**

$x^4 - 5x^2 - 36 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -2\,i, 2\,i$.

Final answer

$x = -3, 3, -2\,i, 2\,i$

Q040 **Written**

$x^4 - 6x^2 - 27 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$.

Final answer

$x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$

Worked Solutions **Higher-Degree Equations**

Q041 Written

$x^4 - x^2 - 2 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -\sqrt{2}, \sqrt{2}, -i,\ i$.

Final answer

$x = -\sqrt{2}, \sqrt{2}, -i,\ i$

Q042 MCQ

$x^4 - 7x^2 - 18 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$.

Correct option: D

Final answer

$x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$

Worked Solutions **Higher-Degree Equations**

Q043 *MCQ*

$x^4 - 5x^2 + 4 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, -1, 1, 2$.

Correct option: D

Final answer

$x = -2, -1, 1, 2$

Q044 *MCQ*

$x^4 - 5x^2 - 36 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -2i, 2i$.

Correct option: C

Final answer

$x = -3, 3, -2i, 2i$

Worked Solutions **Higher-Degree Equations**

Q045 MCQ

$x^4 - 6x^2 - 27 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$.

Correct option: C

Final answer

$x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$

Q046 MCQ

$x^4 - x^2 - 2 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -\sqrt{2}, \sqrt{2}, -i,\ i$.

Correct option: B

Final answer

$x = -\sqrt{2}, \sqrt{2}, -i,\ i$

Worked Solutions **Higher-Degree Equations**

Q047 Written

$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Final answer

$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$

Q048 Written

$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3, 4$.

Final answer

$x = 1, 2, 3, 4$

Worked Solutions **Higher-Degree Equations**

Q049 **Written**

$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3(\text{repeated})$.

Final answer

$x = 1, 2, 3(\text{repeated})$

Q050 **Written**

$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 3, 1 - i, 1 + i$.

Final answer

$x = -1, 3, 1 - i, 1 + i$

Worked Solutions **Higher-Degree Equations**

Q051 MCQ

$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Correct option: B

Final answer

$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$

Worked Solutions **Higher-Degree Equations**

Q052 MCQ

$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1, 2, 3, 4$.

Correct option: A

Final answer

$x = 1, 2, 3, 4$

Q053 MCQ

$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1, 2, 3(\text{repeated})$.

Correct option: C

Final answer

$x = 1, 2, 3(\text{repeated})$

Worked Solutions **Higher-Degree Equations**

Q054 MCQ

$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 3, 1 - i, 1 + i$.

Correct option: C

Final answer

$x = -1, 3, 1 - i, 1 + i$

Q055 Written

$x^3 + 4x^2 + ax + b = 0$ **has roots -3 and 1; find a,b and the other root.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 1, b = -6;$ other roots: -2.

Final answer

$a = 1, b = -6;$ other roots: -2

Worked Solutions **Higher-Degree Equations**

Q056 **Written**

$x^3 - ax^2 - x + b = 0$ **has roots -1 and 2; find a,b and the other root.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 2, b = 2;$ other roots: 1.

Final answer

$a = 2, b = 2;$ other roots: 1

Q057 **Written**

$x^3 + ax^2 + bx - 8 = 0$ **has roots 1 and 2; find a,b and the other root.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -7, b = 14;$ other roots: 4.

Final answer

$a = -7, b = 14;$ other roots: 4

Worked Solutions **Higher-Degree Equations**

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ **has roots -3 and 1; find a,b and the other root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 1, b = -6;\ \text{other roots:}\ -2$.

Correct option: C

Final answer

$a = 1, b = -6;\ \text{other roots:}\ -2$

Worked Solutions **Higher-Degree Equations**

Q059 **MCQ**

$x^3 - ax^2 - x + b = 0$ **has roots -1 and 2; find a,b and the other root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 2, b = 2;$ other roots: 1.

Correct option: B

Final answer

$a = 2, b = 2;$ other roots: 1

Worked Solutions **Higher-Degree Equations**

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ **has roots 1 and 2; find a,b and the other root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -7, b = 14;$ other roots: 4.

Correct option: C

Final answer

$a = -7, b = 14;$ other roots: 4

Worked Solutions **Higher-Degree Equations**

Q061 Written

$x^3 + ax^2 + bx + 5 = 0$ **has root 2+i; find real a,b and the other roots.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -3, b = 1;$ other roots: $-1, 2 - i$.

Final answer

$a = -3, b = 1;$ other roots: $-1, 2 - i$

Q062 Written

$x^3 + ax^2 + bx - 4 = 0$ **has root 1+i; find a,b and the other roots.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 6;$ other roots: $1 - i, 2$.

Final answer

$a = -4, b = 6;$ other roots: $1 - i, 2$

Worked Solutions **Higher-Degree Equations**

Q063 **Written**

$x^3 + ax^2 + bx - 20 = 0$ **has root** $1 - 3\,i$**; find real a,b and the other roots.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 14;$ other roots: $1 + 3i, 2$.

Final answer

$a = -4, b = 14;$ other roots: $1 + 3i, 2$

Q064 **Written**

$x^3 + ax^2 + bx + 10 = 0$ **has root** $1 + 2\,i$**; find real a,b and the other roots.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 0, b = 1;$ other roots: $1 - 2i, -2$.

Final answer

$a = 0, b = 1;$ other roots: $1 - 2i, -2$

Worked Solutions **Higher-Degree Equations**

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ **has root 2+i; find real a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -3, b = 1;$ other roots: $-1, 2 - i$.

Correct option: A

Final answer

$a = -3, b = 1;$ other roots: $-1, 2 - i$

Q066 MCQ

$x^3 + ax^2 + bx - 4 = 0$ **has root 1+i; find a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4, b = 6;$ other roots: $1 - i, 2$.

Correct option: B

Final answer

$a = -4, b = 6;$ other roots: $1 - i, 2$

Worked Solutions **Higher-Degree Equations**

Q067 MCQ

$x^3 + ax^2 + bx - 20 = 0$ **has root** $1 - 3\,i$**; find real a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4, b = 14;$ other roots: $1 + 3i, 2$.

Correct option: D

Final answer

$a = -4, b = 14;$ other roots: $1 + 3i, 2$

Q068 MCQ

$x^3 + ax^2 + bx + 10 = 0$ **has root** $1 + 2\,i$**; find real a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 0, b = 1;$ other roots: $1 - 2i, -2$.

Correct option: B

Final answer

$a = 0, b = 1;$ other roots: $1 - 2i, -2$

Worked Solutions **Higher-Degree Equations**

Q069 Written

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ **has root 2; find k and the other roots.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $k = 9;$ other roots: $3, 4$.

Final answer

$k = 9;$ other roots: $3, 4$

Q070 MCQ

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ **has root 2; find k and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $k = 9;$ other roots: $3, 4$.

Correct option: B

Final answer

$k = 9;$ other roots: $3, 4$

Worked Solutions **Higher-Degree Equations**

Q071 Written

For an imaginary cube root of unity w, evaluate w^9.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q072 Written

For an imaginary cube root of unity w, evaluate w^{2010}.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions **Higher-Degree Equations**

Q073 Written

For an imaginary cube root of unity w, evaluate w^3.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q074 MCQ

For an imaginary cube root of unity w, evaluate w^9 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: A

Final answer

1

Worked Solutions **Higher-Degree Equations**

Q075 MCQ

For an imaginary cube root of unity w, evaluate w^{2010} Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: B

Final answer

1

Q076 MCQ

For an imaginary cube root of unity w, evaluate w^3 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: D

Final answer

1

Worked Solutions **Higher-Degree Equations**

Q077 **Written**

For an imaginary cube root of unity w, evaluate $w^4 + w^5$.

Solution

1. Use $(w^3 = 1)$ and reduce powers modulo 3.
2. Use $(1 + w + w^2 = 0)$ to simplify the remaining expression.
3. Therefore the final answer is -1.

Final answer

-1

Q078 **Written**

For an imaginary cube root of unity w, evaluate $w^2 + w$.

Solution

1. Use $(w^3 = 1)$ and reduce powers modulo 3.
2. Use $(1 + w + w^2 = 0)$ to simplify the remaining expression.
3. Therefore the final answer is -1.

Final answer

-1

Worked Solutions **Higher-Degree Equations**

Q079 Written

For an imaginary cube root of unity w, evaluate $w^2 + w^4 + w^6$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q080 MCQ

For an imaginary cube root of unity w, evaluate $w^4 + w^5$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1.

Correct option: A

Final answer

-1

Worked Solutions **Higher-Degree Equations**

Q081 MCQ

For an imaginary cube root of unity w, evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1.

Correct option: B

Final answer

-1

Q082 MCQ

For an imaginary cube root of unity w, evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 0.

Correct option: D

Final answer

0

Worked Solutions **Higher-Degree Equations**

Q083 Written

For an imaginary cube root of unity w, evaluate $w + 1/w$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1.

Final answer

-1

Q084 Written

For an imaginary cube root of unity w, evaluate $(1 + w)(1 + w^2)$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions **Higher-Degree Equations**

Q085 **Written**

For an imaginary cube root of unity w, evaluate $1/w + 1/w^2 + 1$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q086 **MCQ**

For an imaginary cube root of unity w, evaluate $w + 1/w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1.

Correct option: D

Final answer

-1

Worked Solutions **Higher-Degree Equations**

Q087 MCQ

For an imaginary cube root of unity w, evaluate $(1+w)(1+w^2)$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: C

Final answer

1

Q088 MCQ

For an imaginary cube root of unity w, evaluate $1/w + 1/w^2 + 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 0.

Correct option: D

Final answer

0

Worked Solutions **Higher-Degree Equations**

Q089 **Written**

$x^3 - 2x^2 + (a-3)x + a = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -4$ or $9/4$.

Final answer

$a = -4$ or $9/4$

Q090 **Written**

$x^3 - x^2 + (a-2)x + a = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -3$ or 1.

Final answer

$a = -3$ or 1

Worked Solutions **Higher-Degree Equations**

Q091 **Written**

$x^3 - 7x^2 + (a+6)x - a = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 5$ or 9.

Final answer

$a = 5$ or 9

Q092 **Written**

$x^3 - (a+1)x^2 + (3a+5)x - 2a - 5 = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -6, -2,$ or 10.

Final answer

$a = -6, -2,$ or 10

Worked Solutions **Higher-Degree Equations**

Q093 Written

$x^3 + (2a-1)x^2 - 3(a-2)x + a - 6 = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose $(D = 0)$, then retain the real parameter values.
3. Therefore the final answer is $a = -7, -3,$ or 2.

Final answer

$a = -7, -3,$ or 2

Q094 MCQ

$x^3 - 2x^2 + (a-3)x + a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4$ or $9/4$.

Correct option: D

Final answer

$a = -4$ or $9/4$

Worked Solutions **Higher-Degree Equations**

Q095 MCQ

$x^3 - x^2 + (a-2)x + a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -3$ or 1.

Correct option: D

Final answer

$a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a+6)x - a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 5$ or 9.

Correct option: B

Final answer

$a = 5$ or 9

Worked Solutions **Higher-Degree Equations**

Q097 MCQ

$x^3-(a+1)x^2+(3a+5)x-2a-5=0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a=-6,-2,\ \text{or } 10$.

Correct option: A

Final answer

$a=-6,-2,\ \text{or } 10$

Q098 MCQ

$x^3+(2a-1)x^2-3(a-2)x+a-6=0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a=-7,-3,\ \text{or } 2$.

Correct option: A

Final answer

$a=-7,-3,\ \text{or } 2$

Worked Solutions **Higher-Degree Equations**

Q099 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}\,i$; find real a,b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$.

Final answer

$a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$

Worked Solutions **Higher-Degree Equations**

Q100 **Challenge**

For an imaginary cube root of unity w, evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions **Higher-Degree Equations**

Q101 Challenge

$x^3 + (a-1)x^2 + (a-3)x - 2a + 3 = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose $(D = 0)$, then retain the real parameter values.
3. Therefore the final answer is $a = 2/3, 2,\ \text{or } 6$.

Final answer

$a = 2/3, 2,\ \text{or } 6$

Worked Solutions **Higher-Degree Equations**

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Solution

1. Set inflow equal to outflow: $(x^4 + x^2 + x = x^3 + 2)$.
2. Move all terms to one side.
3. Therefore the final answer is $x^4 - x^3 + x^2 + x - 2 = 0$.

Final answer

$x^4 - x^3 + x^2 + x - 2 = 0$

Worked Solutions **Higher-Degree Equations**

Q103 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels.

Solution

1. Factor $(x^4 - x^3 + x^2 + x - 2)$ and solve the remaining quadratic.
2. Apply the stated physical restriction (xge0) only after listing the algebraic roots.
3. Therefore the final answer is $x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$.

Final answer

$x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$

Q104 Challenge

Classify the stable pressure solutions as real or complex.

Solution

1. Separate the real roots from the non-real conjugate pair.

$$\text{Both types occur: real } x = -1, 1$$

2. Therefore the final answer is $\text{non-real conjugates } x = (1 \pm \sqrt{7}\, i)/2$.

$$\text{if } x \geq 0 : \quad x = 1$$

Final answer

$$\text{Both types occur: real } x = -1, 1$$
$$\text{non-real conjugates } x = (1 \pm \sqrt{7}\, i)/2$$
$$\text{if } x \geq 0 : \quad x = 1$$

Worked Solutions Higher-Degree Equations

Q105 **Challenge**

$x^3 + ax + b = 0$ **has root** $1 + \sqrt{2}\,i$**; find real a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$.

Correct option: C

Final answer

$a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$

Worked Solutions **Higher-Degree Equations**

Q106 Challenge

For an imaginary cube root of unity w, evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 4.

Correct option: D

Final answer

4

Worked Solutions **Higher-Degree Equations**

Q107 **Challenge**

$x^3 + (a-1)x^2 + (a-3)x - 2a + 3 = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 2/3, 2,$ or 6.

Correct option: D

Final answer

$a = 2/3, 2,$ or 6

Worked Solutions **Higher-Degree Equations**

Q108 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x^4 - x^3 + x^2 + x - 2 = 0$.

Correct option: A

Final answer

$x^4 - x^3 + x^2 + x - 2 = 0$

Worked Solutions **Higher-Degree Equations**

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$.

Correct option: D

Final answer

$x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q110 Challenge

Classify the stable pressure solutions as real or complex Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.

$$\text{Both types occur: real } x=-1,1$$

2. The correct value is non-real conjugates $x=(1\pm\sqrt{7}\,i)/2$.

$$\text{if } x\ge 0:\ \ x=1$$

Correct option: B

Final answer

Both types occur: real $x=-1,1$

non-real conjugates $x=(1\pm\sqrt{7}\,i)/2$

$$\text{if } x\ge 0:\ \ x=1$$

Worked Solutions **Higher-Degree Equations**

Quiz MCQ 1 Concept Quiz · MCQ

$Solve x^3 - 8 = 0$. **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$.

Correct option: D

Final answer

$x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

Quiz MCQ 2 Concept Quiz · MCQ

$A real cubic has root 1 + 2\,i. Which root must also occur$? **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $1 - 2\,i$.

Correct option: D

Final answer

$1 - 2\,i$

Worked Solutions **Higher-Degree Equations**

Quiz MCQ 3 **Concept Quiz · MCQ**

For an imaginary cube root of unity w, evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1.

Correct option: B

Final answer

-1

Quiz MCQ 4 **Concept Quiz · MCQ**

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $D = 0$.

Correct option: B

Final answer

$D = 0$

Worked Solutions **Higher-Degree Equations**

Quiz WRITTEN 5 **Concept Quiz · Written**

$Solve x^4 - 5x^2 + 4 = 0.$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $x = -2, -1, 1, 2$

Final answer

$x = -2, -1, 1, 2$

Quiz WRITTEN 6 **Concept Quiz · Written**

$For x^3 + ax^2 + bx + 10 = 0$ **with root** $1 + 2\,i$**, find a,b and the other roots.**

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = 0, b = 1;$ other roots: $1 - 2i, -2$

Final answer

$a = 0, b = 1;$ other roots: $1 - 2i, -2$

Worked Solutions **Higher-Degree Equations**

Quiz WRITTEN 7 **Concept Quiz · Written**

For an imaginary cube root of unity w, evaluate $(1 + w)(1 + w^2)$.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: 1

Final answer

1

Quiz WRITTEN 8 **Concept Quiz · Written**

$Findallrealaforx^3 - 2x^2 + (a - 3)x + a = 0tohavearepeatedroot.$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = -4$ or $9/4$

Final answer

$a = -4$ or $9/4$

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1-\sqrt{3}\,i, -1+\sqrt{3}\,i$

Q002 $x = -1, (1-\sqrt{3}\,i)/2, (1+\sqrt{3}\,i)/2$

Q003 $x = 3, (-3-3\sqrt{3}\,i)/2, (-3+3\sqrt{3}\,i)/2$

Q004 $x = -5, (5-5\sqrt{3}\,i)/2, (5+5\sqrt{3}\,i)/2$

Q005 $x = 1/2, (-1-\sqrt{3}\,i)/4, (-1+\sqrt{3}\,i)/4$

Q006 B

Q007 D

Q008 B

Q009 A

Q010 C

Q011 $x = -2, 1, 3$

Q012 $x = -2, 2/3, 1$

Q013 $x = 2, (-1-\sqrt{7}\,i)/4, (-1+\sqrt{7}\,i)/4$

Q014 $x = -2(\text{repeated}), 5$

Q015 $x = -3, -2, -1$

Q016 $x = -3, -2, 2$

Q017 $x = -1, 2/3, 1$

Q018 $x = 2, (-1-\sqrt{7}\,i)/2, (-1+\sqrt{7}\,i)/2$

Q019 $x = -2, -1-\sqrt{3}, -1+\sqrt{3}$

Q020 $x = 2, 1-2\,i, 1+2\,i$

Q021 $x = -2(\text{repeated}), 3$

Q022 $x = -1(\text{repeated}), -3/2$

Q023 $x = -2(\text{triple})$

Q024 C

Q025 D

Q026 D

Q027 B

Q028 C

Q029 D

Q030 D

Q031 A

Q032 D

Q033 D

Q034 D

Q035 B

Q036 D

Q037 $x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$

Q038 $x = -2, -1, 1, 2$

Q039 $x = -3, 3, -2\,i, 2\,i$

Q040 $x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$

Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$

Q042 D

Q043 D

Q044 C

Q045 C

Q046 B

Q047 $x = 2(\text{repeated}), 1-\sqrt{5}, 1+\sqrt{5}$

Q048 $x = 1, 2, 3, 4$

Q049 $x = 1, 2, 3(\text{repeated})$

Q050 $x = -1, 3, 1-i, 1+i$

Q051 B

Q052 A

Q053 C

Q054 C

Q055 $a = 1, b = -6;$ other roots: -2

Q056 $a = 2, b = 2;$ other roots: 1

Q057 $a = -7, b = 14;$ other roots: 4

Q058 C

Q059 B

Q060 C

Q061 $a = -3, b = 1;$ other roots: $-1, 2-i$

Q062 $a = -4, b = 6;$ other roots: $1-i, 2$

Q063 $a = -4, b = 14;$ other roots: $1+3i, 2$

Q064 $a = 0, b = 1;$ other roots: $1-2i, -2$

Q065 A

Q066 B

Q067 D

Q068 B

Q069 $k = 9;$ other roots: $3, 4$

Q070 B

Q071 1

Q072 1

Q073 1

Q074 A

Q075 B

Q076 D

Q077 -1

Q078 -1

Q079 0

Q080 A

Q081 B

Q082 D

Q083 -1

Q084 1

Q085 0

Q086 D

Q087 C

Q088 D

Q089 $a = -4$ or $9/4$

Q090 $a = -3$ or 1

Q091 $a = 5$ or 9

Q092 $a = -6, -2,$ or 10

Q093 $a = -7, -3,$ or 2

Q094 D

Q095 D

Q096 B

Q097 A

Q098 A

Q099 $a = -1, b = 6;$ other roots: $1-\sqrt{2}i, -2$

Q100 4

Q101 $a = 2/3, 2,$ or 6

Q102 $x^4 - x^3 + x^2 + x - 2 = 0$

Q103 $x = -1, 1, (1-\sqrt{7}\,i)/2, (1+\sqrt{7}\,i)/2$

Q104 Both types occur: real $x = -1, 1$
non-real conjugates $x = (1 \pm \sqrt{7}\,i)/2$
if $x \geq 0$: $x = 1$

Q105 C

Q106 D

Q107 D

Q108 A

Q109 D

Q110 B

Quiz MCQ 1 D

Quiz MCQ 2 D

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $x = -2, -1, 1, 2$

Quiz WRITTEN 6 $a = 0, b = 1;$ other roots: $1 - 2i, -2$

Quiz WRITTEN 7 1

Quiz WRITTEN 8 $a = -4$ or $9/4$